

고속 인터페이스 CDR 회로의 기초

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Contents

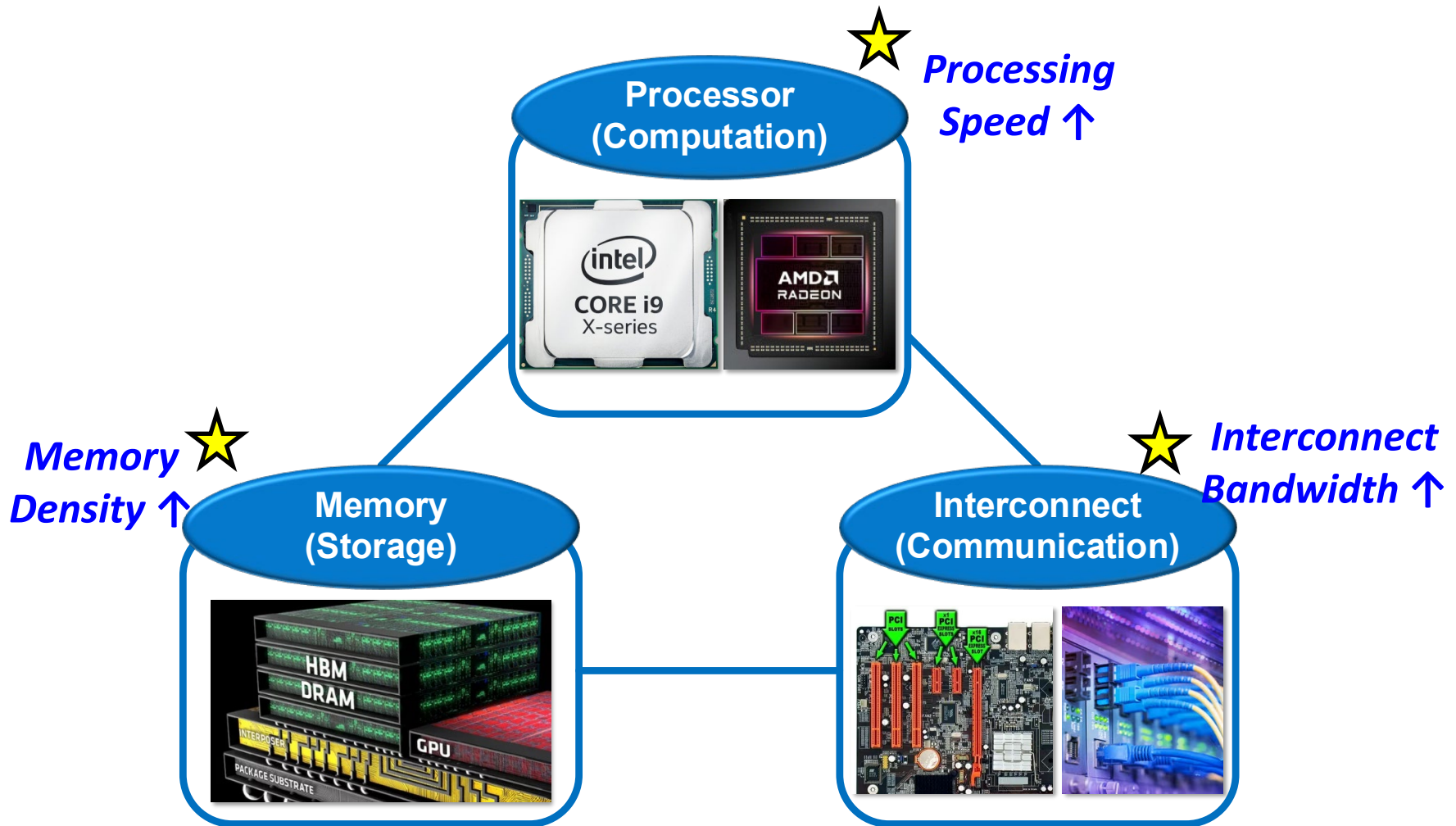
- High-Speed Serial Link
 - Trends for High Performance Computing
 - Applications
 - Bandwidth and Energy Efficiency
 - Architecture Evolutions
- Clock and Data Recovery (CDR)
 - Basics of CDR circuit
 - CDR Architectures
 - CDR Phase Detectors
 - CDR Jitter Metrics

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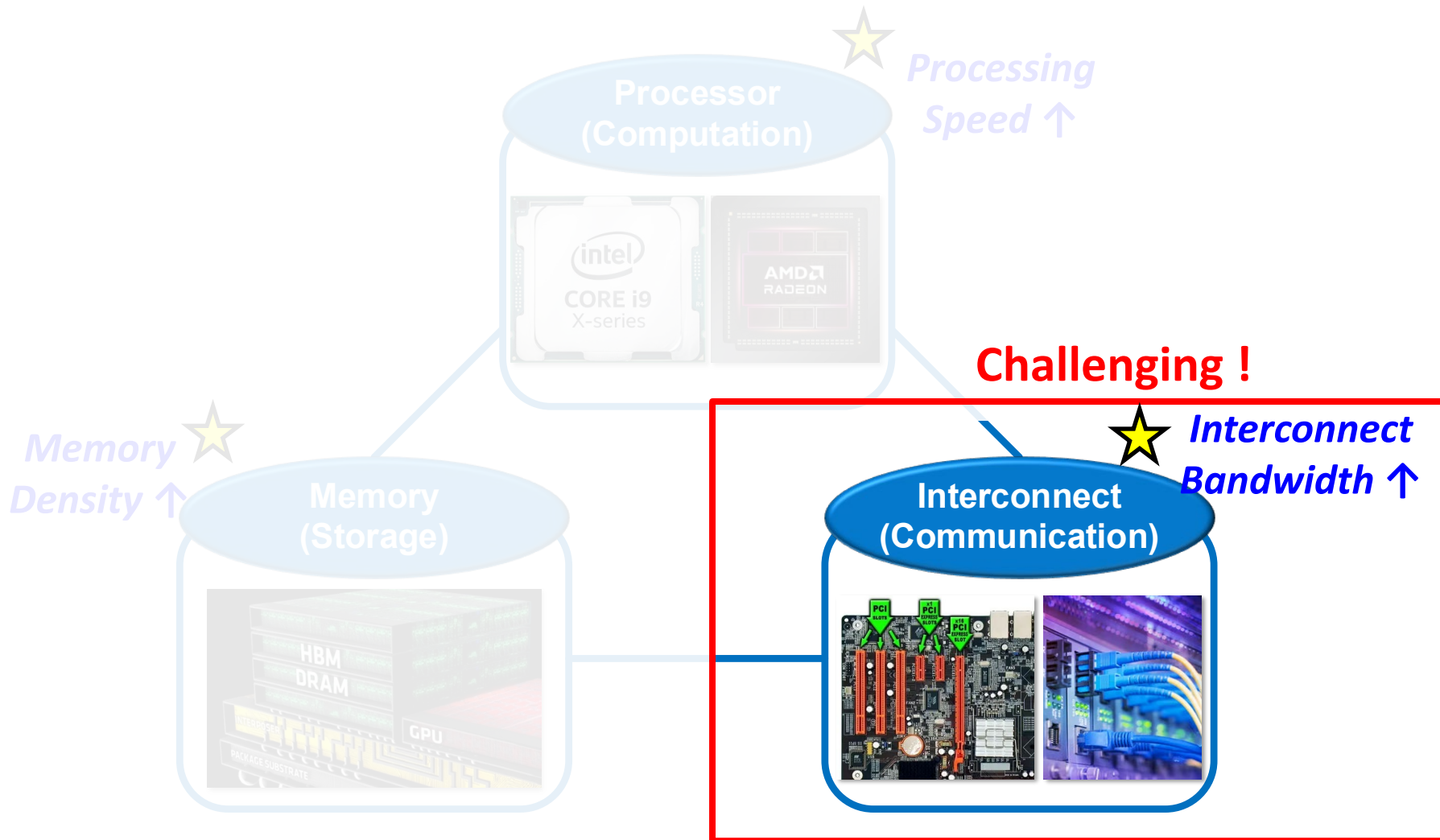
Trends – High Performance Computing (HPC)

- Computing System



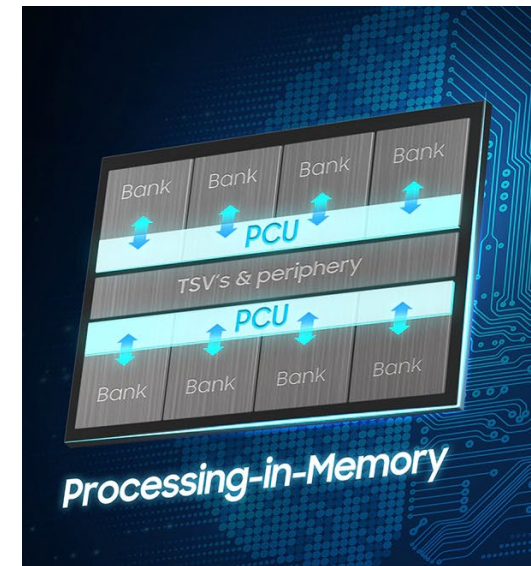
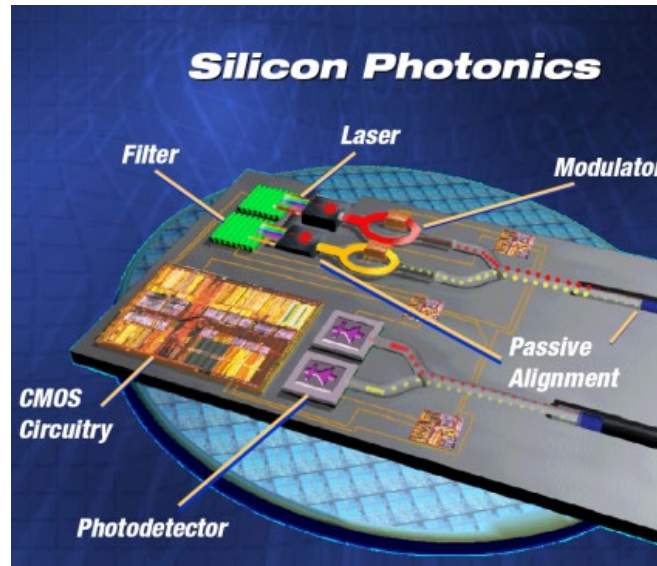
Trends – High Performance Computing (HPC)

- Computing System



Trends – High Performance Computing (HPC)

- Research directions for developing HPC
 - New platforms with system-level innovations
 - System in package (SiP), Chiplet, Silicon Photonics, Processing in Memory (PIM)
 - Specialized high-speed interfaces are required



High-Speed Serial Links

- Why high-speed “serial” link?



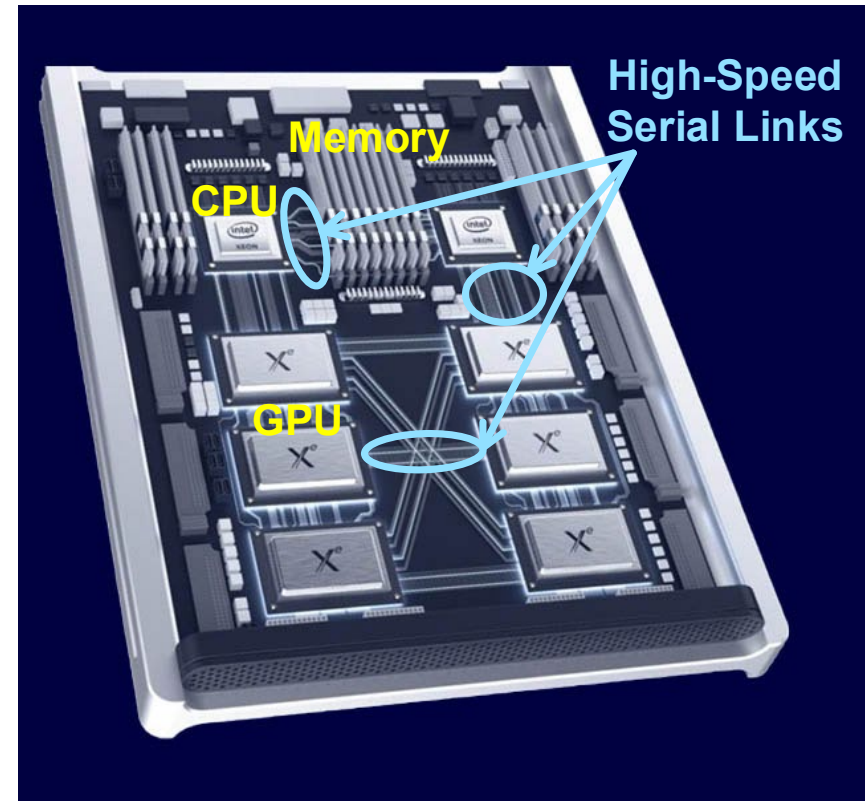
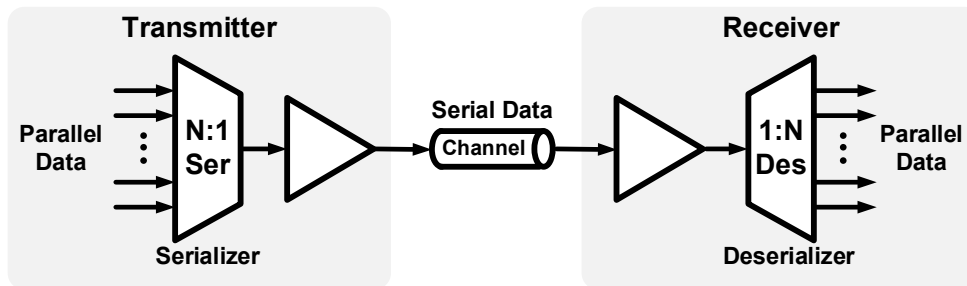
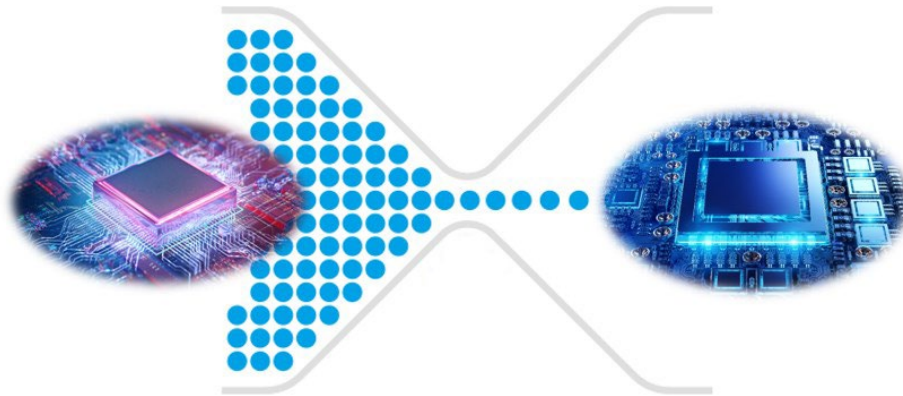
High-Speed Serial Links

- Why high-speed “serial” link?

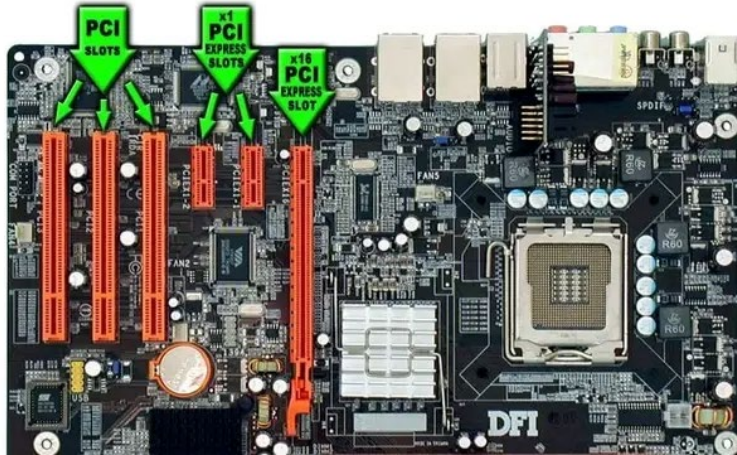


High-Speed Serial Links

- Circuits and systems that transmit/receive high-speed data between integrated circuits
- High-speed I/Os, high-speed interfaces, SerDes...



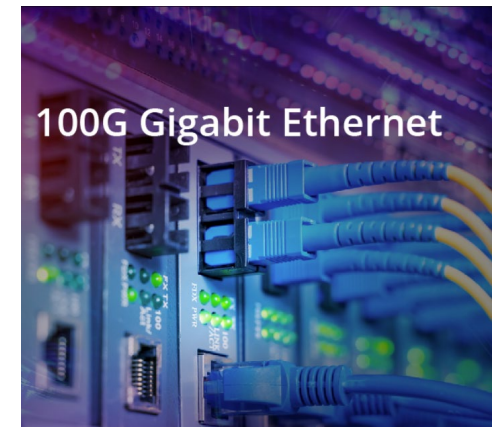
Applications of High-Speed I/O (1)



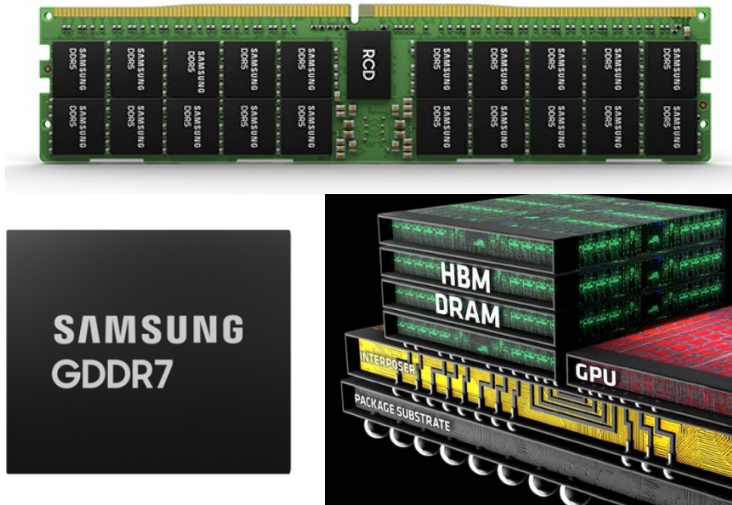
PCI Express



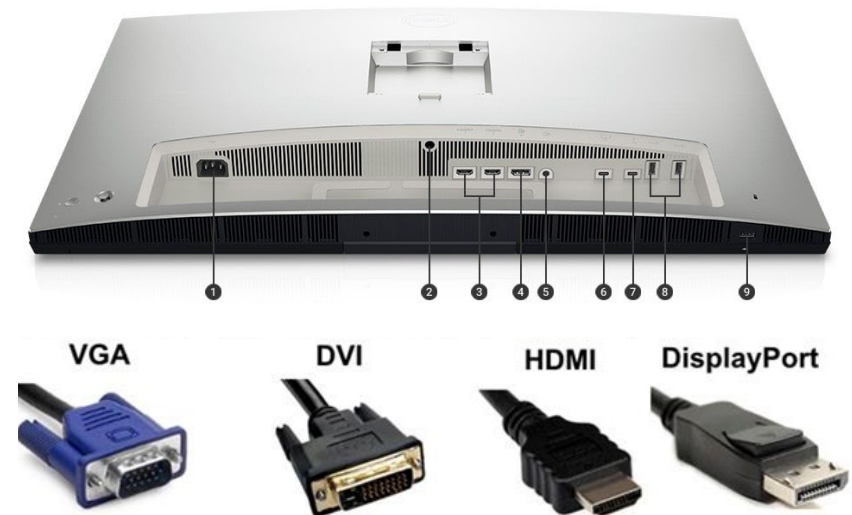
USB



Ethernet



Memory Interfaces

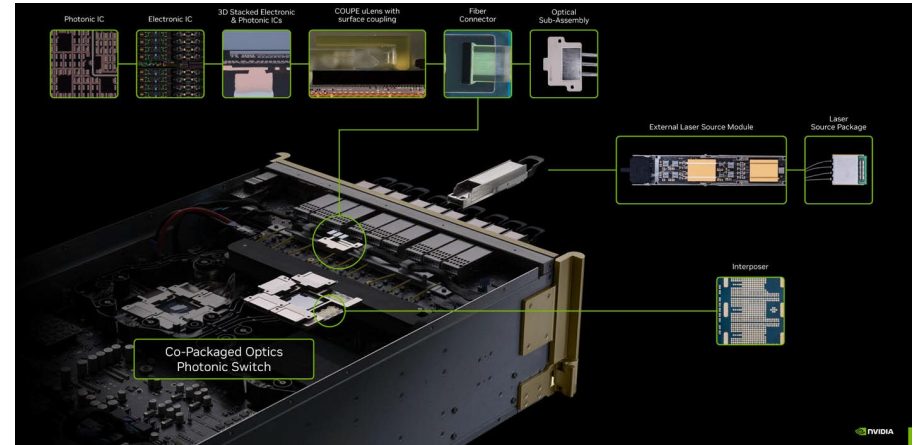


Display Standards

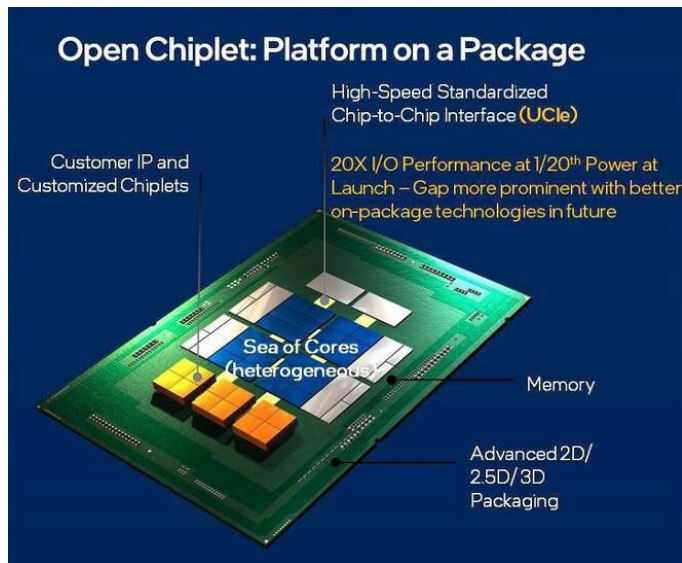
Applications of High-Speed I/O (2)



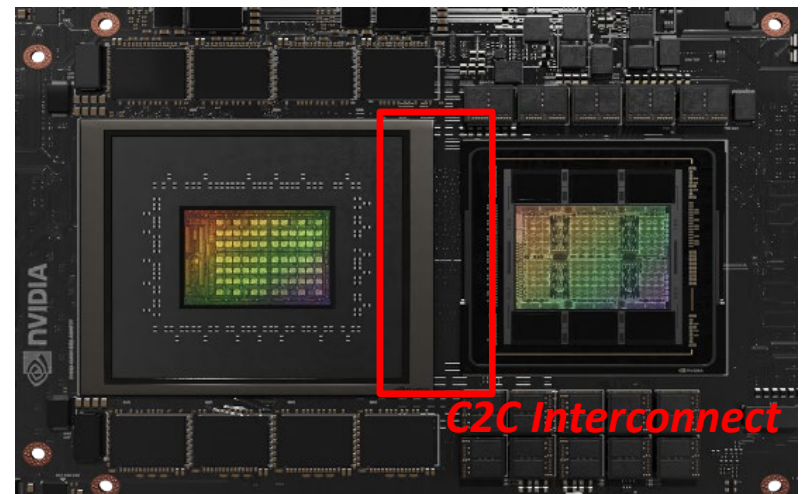
Data Center



Co-packaged Optics



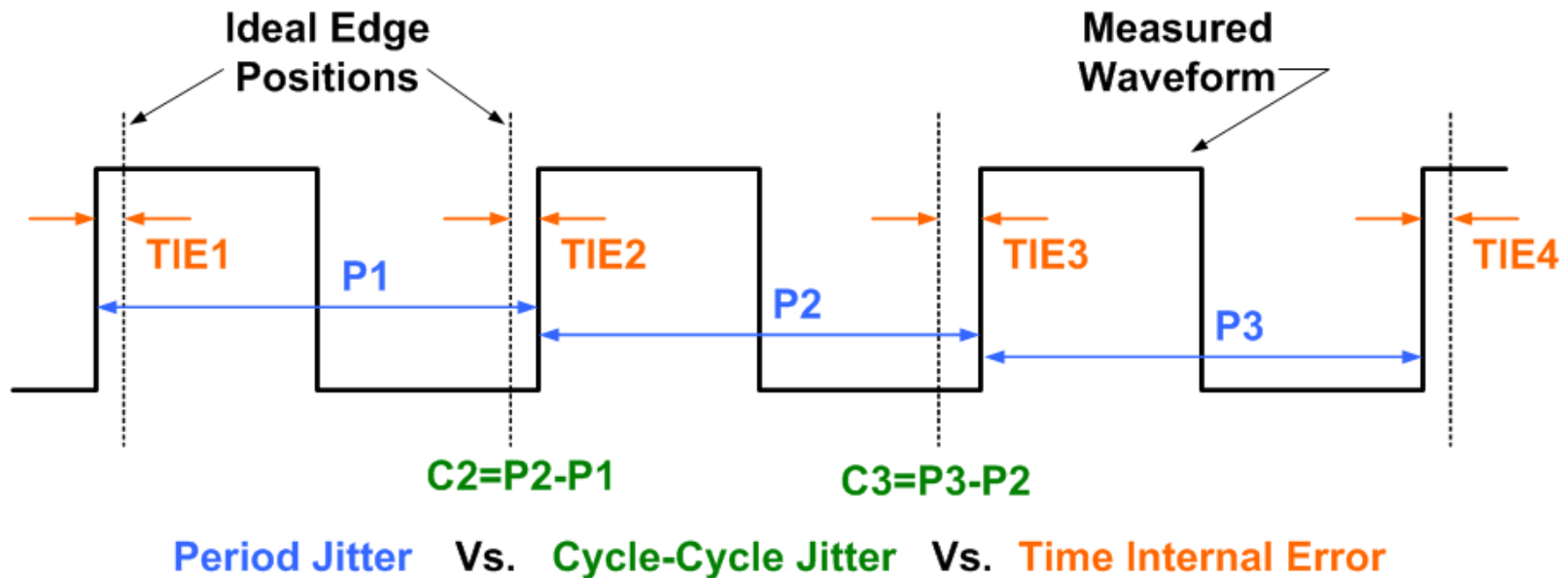
Chiplet, UCIe



NVIDIA Grace Hopper Superchip

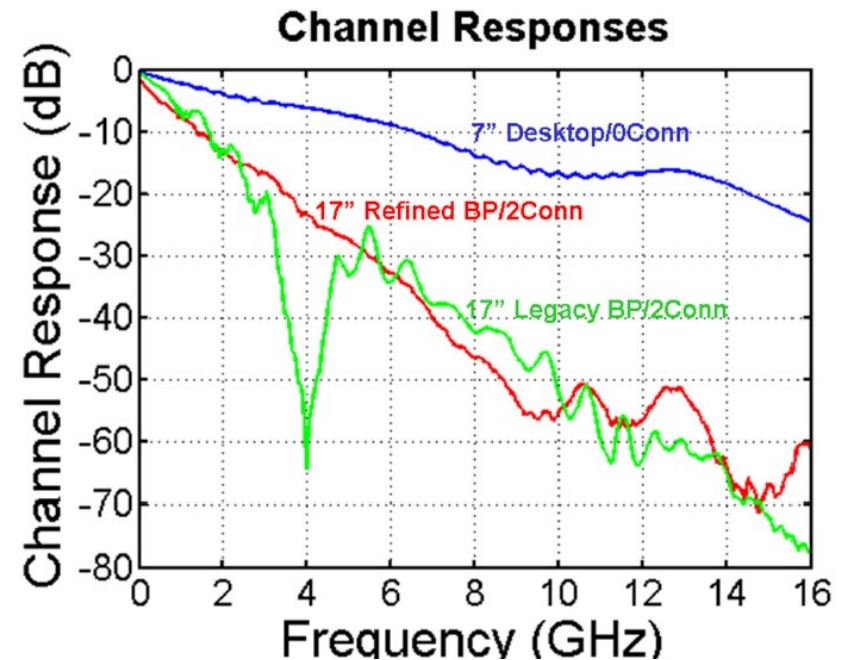
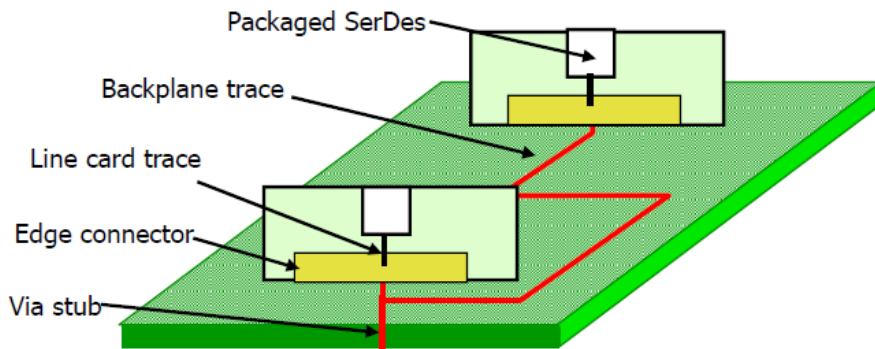
Basic Knowledge of Signaling

- Jitter
 - Definition in SONET specification: “the short term variations of a digital signal’s significant instants from their ideal positions in time”
 - Random jitter and deterministic jitter
- Clock jitter: period jitter, cycle-to-cycle jitter, and time interval error



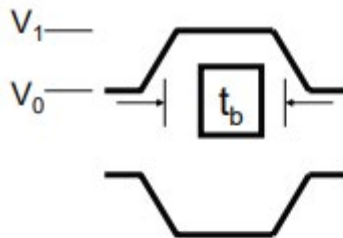
Basic Knowledge of Signaling

- Channel loss
 - Electrical channels basically have low-pass profiles
 - More attenuations and distortions at high frequencies



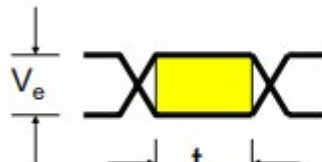
Basic Knowledge of Signaling

- Eye diagram
 - Indicate the 'quality' of received signals
 - Vertical margin and horizontal margin

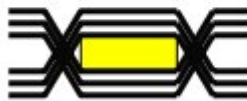


This is a "1"

This is a "0"



Eye Opening - space between 1 and 0

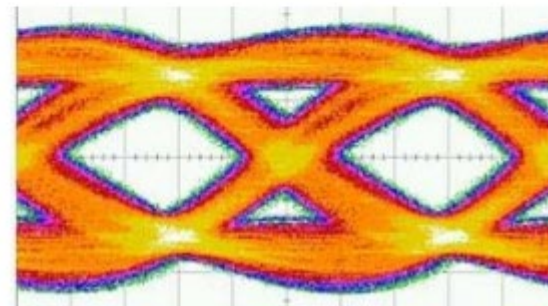


With voltage noise



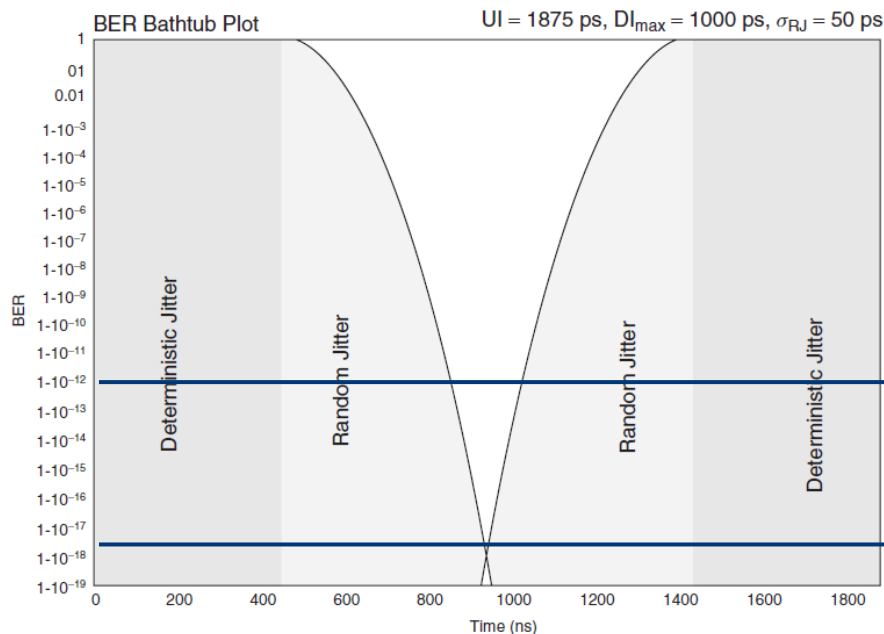
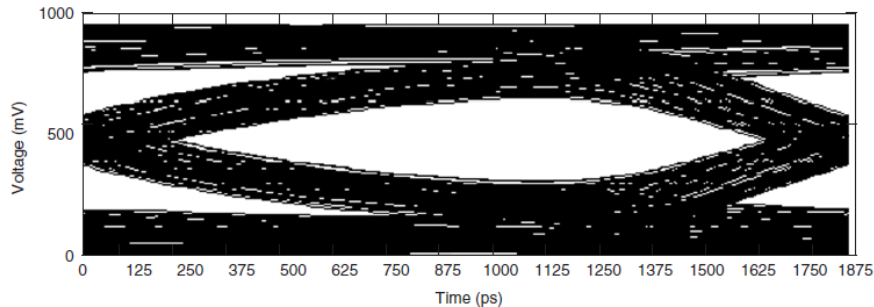
With timing noise

With Both!



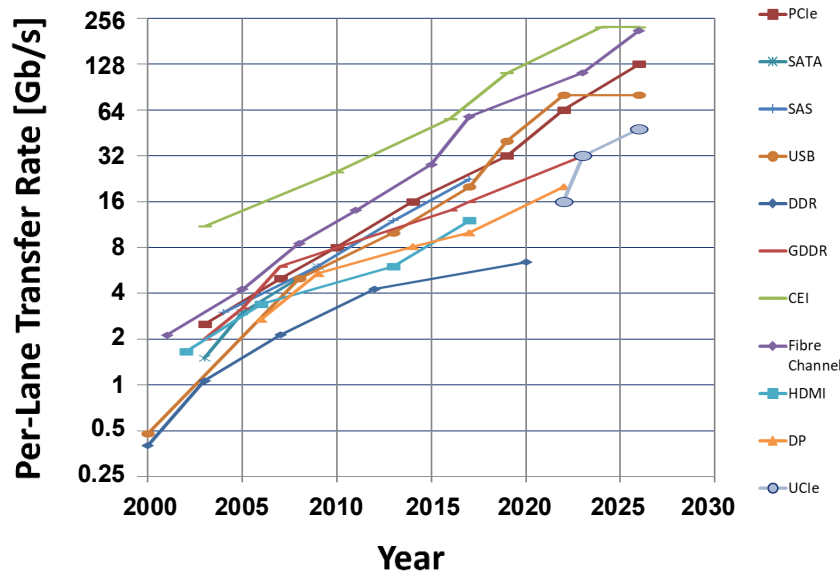
Basic Knowledge of Signaling

- Bit error rate (BER)
 - Average # of wrong received bits / total transmitted bits

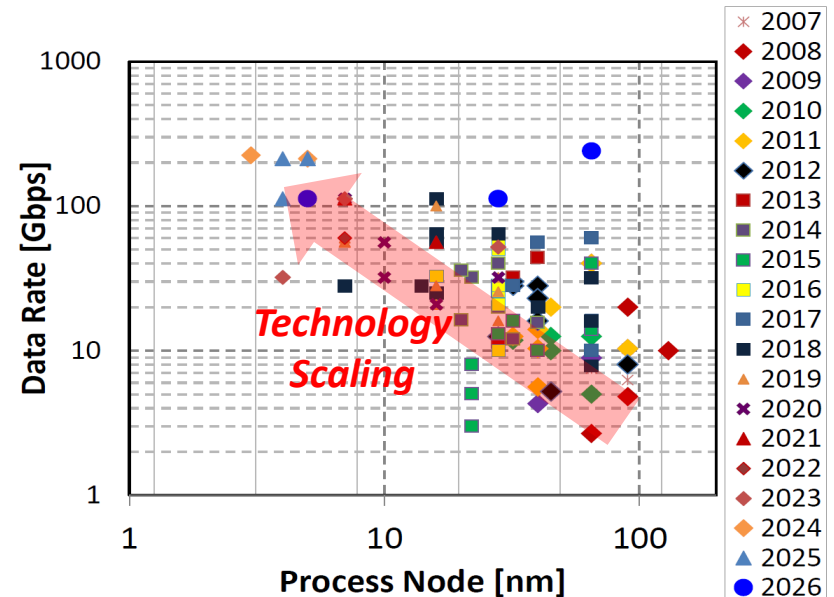


Trends of High-Speed I/O – Bandwidth

- Demands for higher I/O bandwidth
 - Data rate increases by nearly 2x every 4 years
 - **Huge demands for 200+Gb/s data rate**
 - Both advanced circuit techniques and process node scaling contribute to the data rate increase



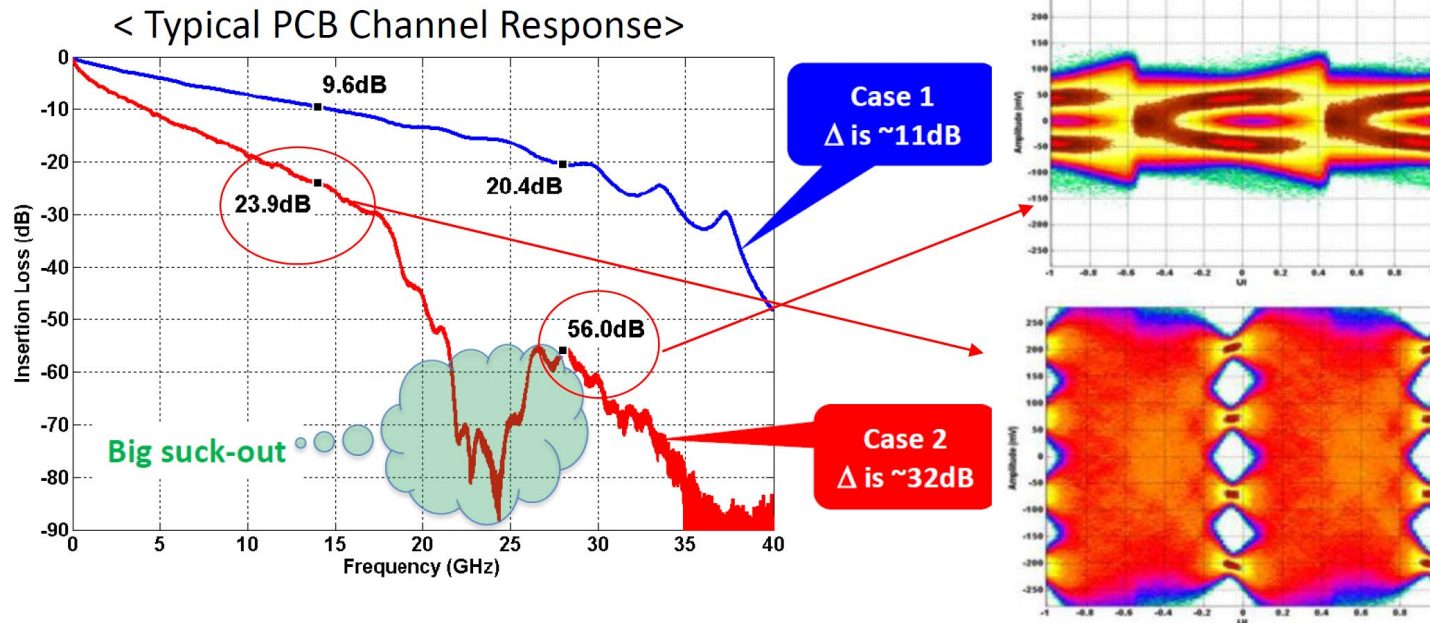
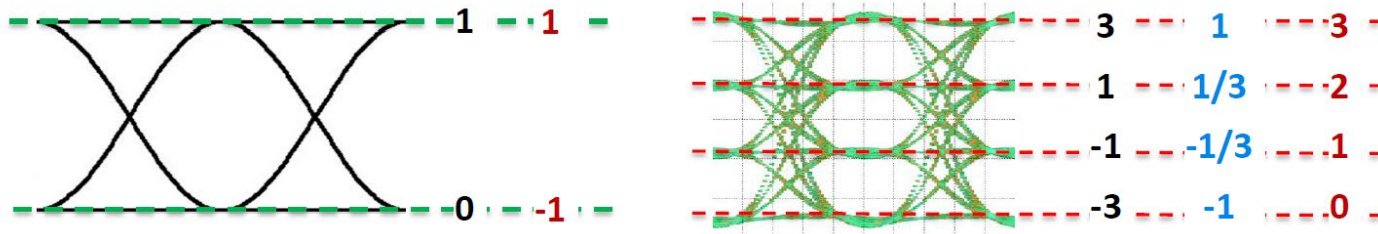
*Per-lane Data Rate vs. Year
for Various I/O Standards*



Data Rate vs. Process Node

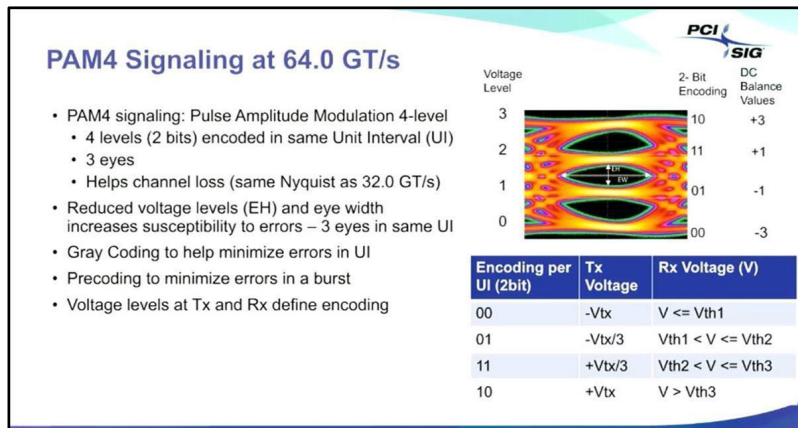
Trends of High-Speed I/O – Bandwidth

- Demands for higher I/O bandwidth
 - **PAM4 modulation** provides better spectral efficiency
 - Many standards are adopting the PAM4 modulation

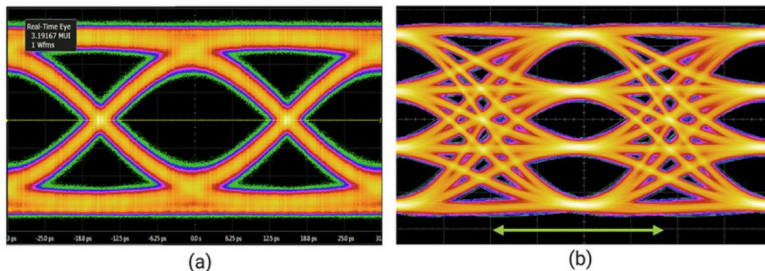


Trends of High-Speed I/O – Bandwidth

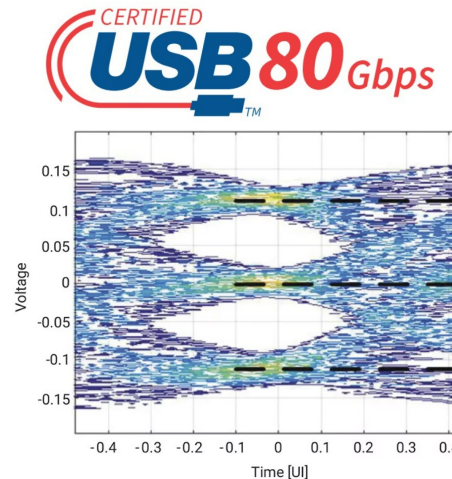
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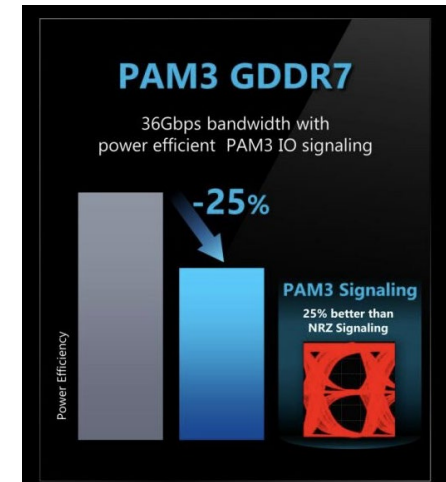
Synopsys PHY IP in TSMC N5
Comparison of 32G NRZ and 64G PAM-4 Eyes



PCI Express 6.0
64Gb/s PAM4



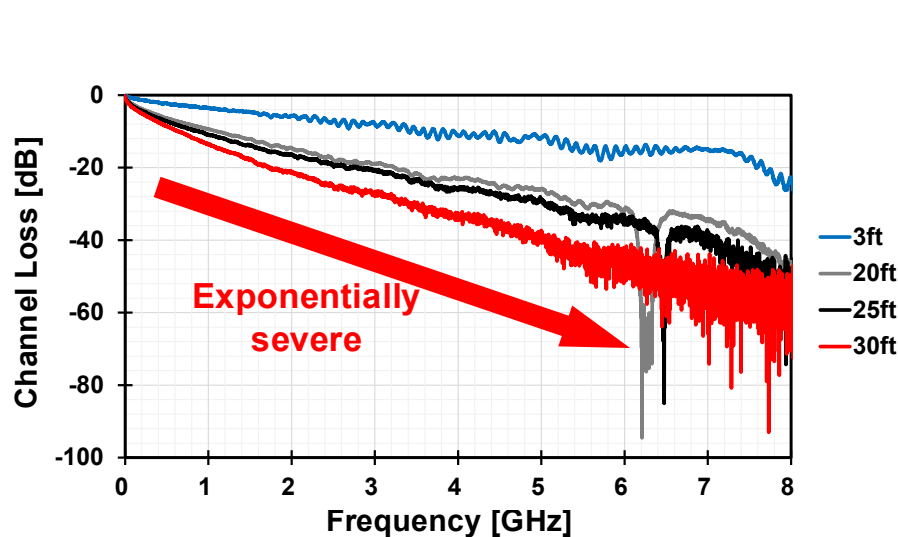
USB4
80Gb/s PAM3



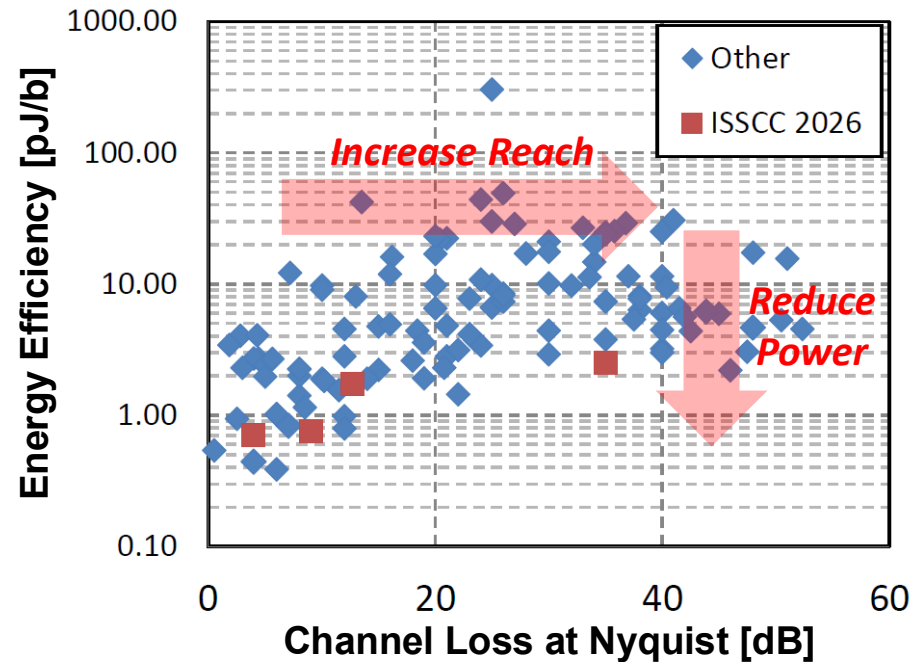
GDDR7
36Gb/s PAM3

Trends of High-Speed I/O – Energy

- Demands for improving energy efficiency
 - Energy efficiency: energy consumed to transfer one bit
 - Strongly coupled to data rate and channel loss



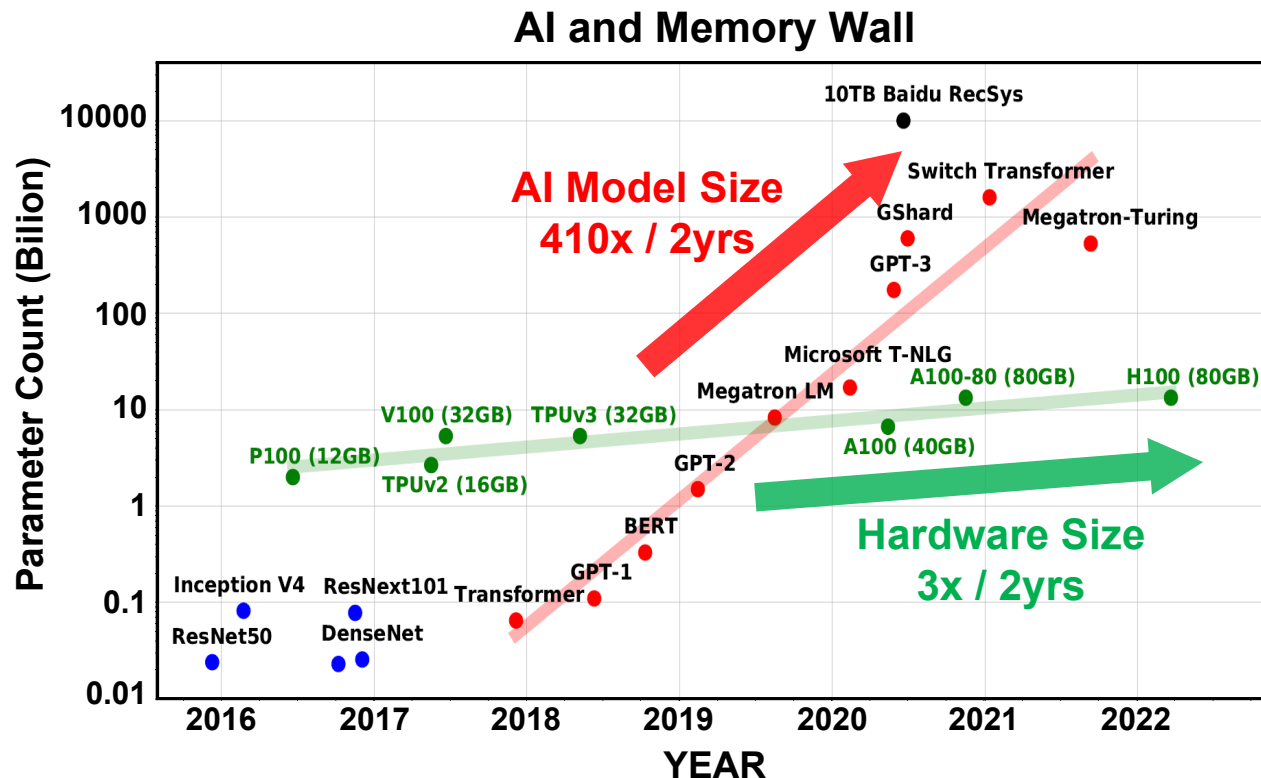
Channel Responses of Various DisplayPort Cables



Energy efficiency vs. Channel Loss

Trends of High-Speed I/O – Throughput

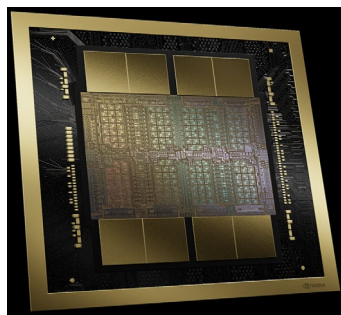
- Demands for higher throughput in large-scale AI hardware systems
 - AI model complexity is increasing much faster than hardware scaling
 - **Hardware scale-up and scale-out** approaches are essential to bridge this gap



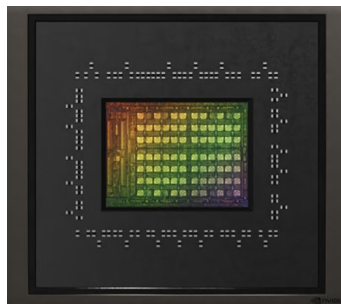
Trends of High-Speed I/O – Throughput

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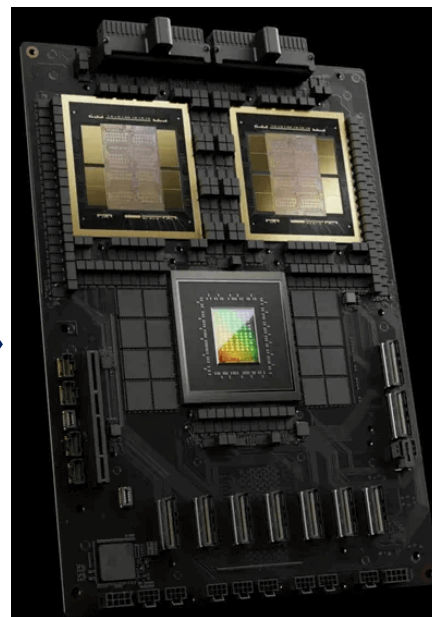
"Scale-Out" of NVIDIA's Blackwell Platform



Blackwell GPU



Grace CPU



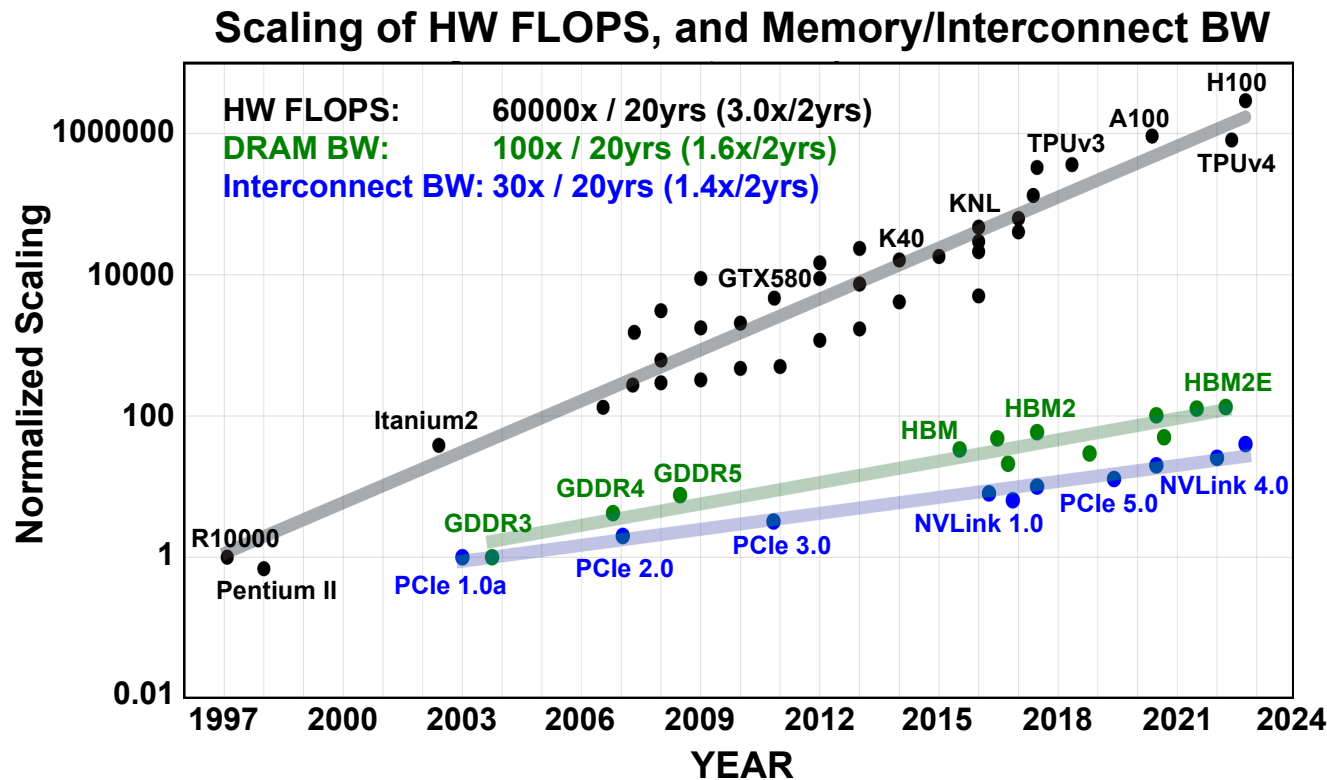
GB200 Superchip
2GPU + 1CPU



GB200 NVL72
Rack-Scale Solution

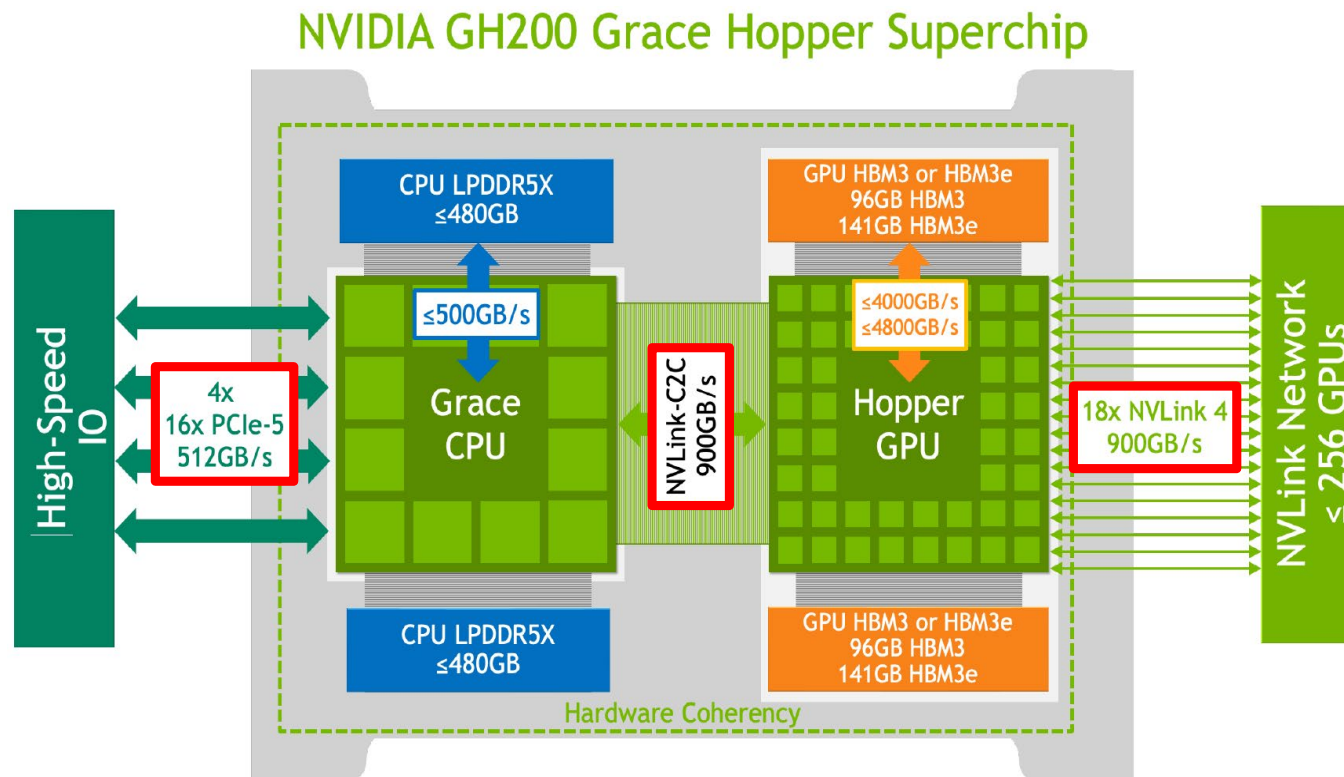
Trends of High-Speed I/O – Throughput

- Hardware scaling faces significant challenge of interconnect bottleneck
- Massive data transmission is strongly required
 - Both high-speed serial I/O and high-density wide I/O are essential



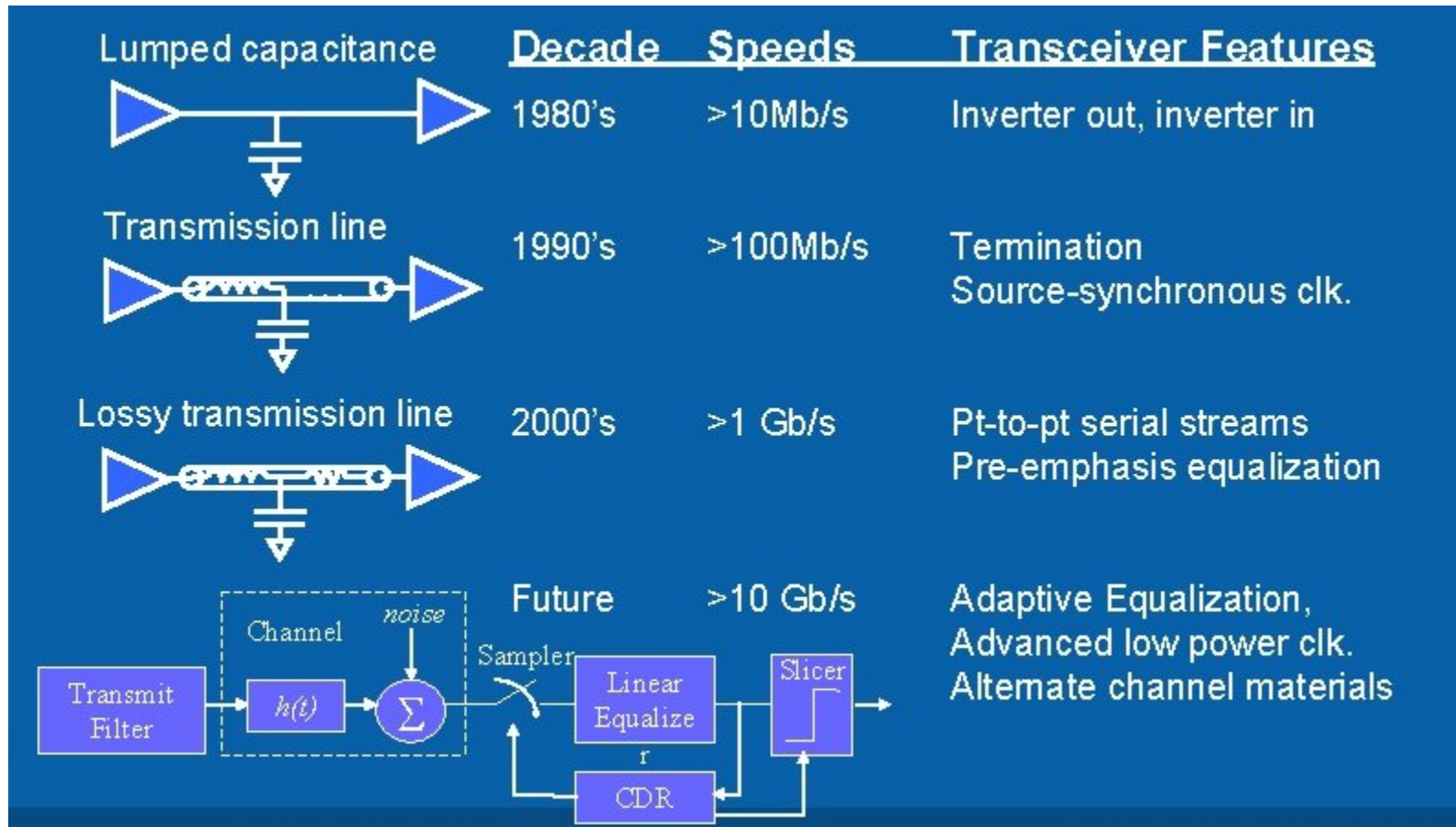
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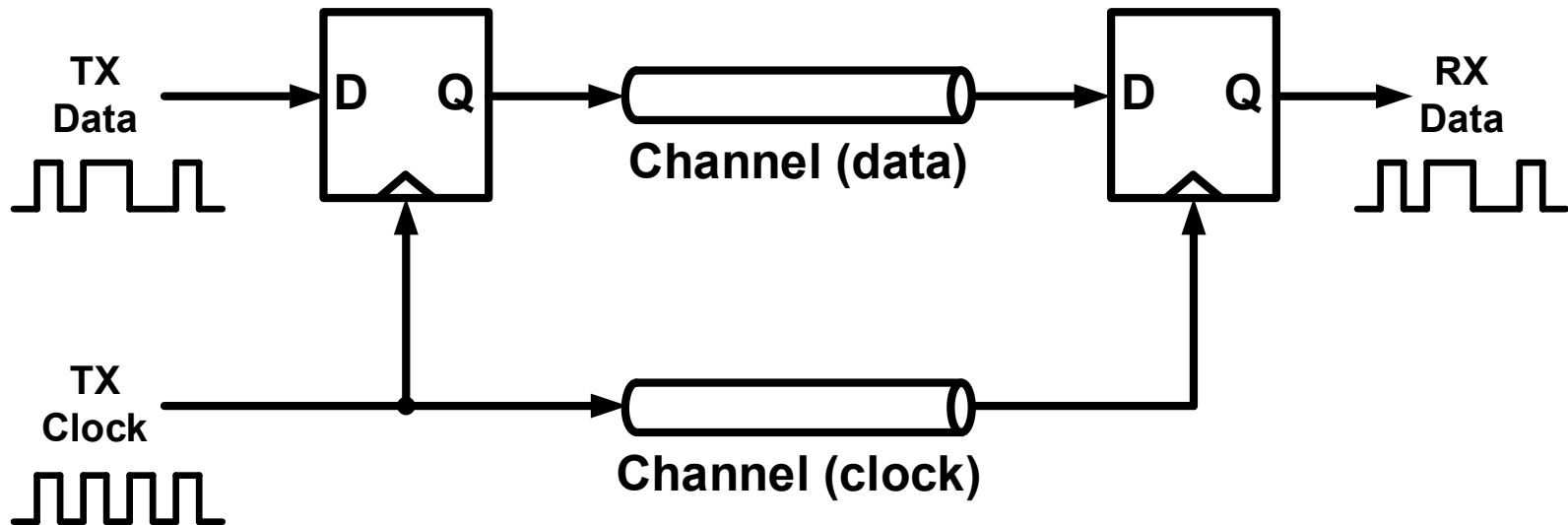
Architecture Evolutions

- Architecture of chip-to-chip communication has been complicated as the data speed increases



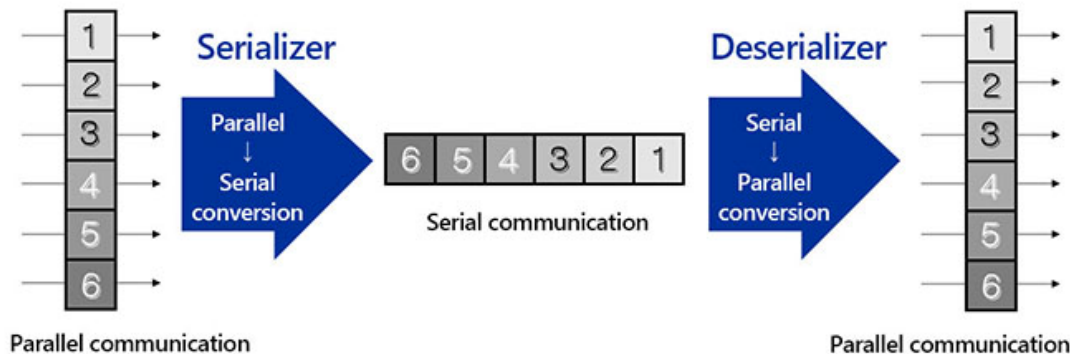
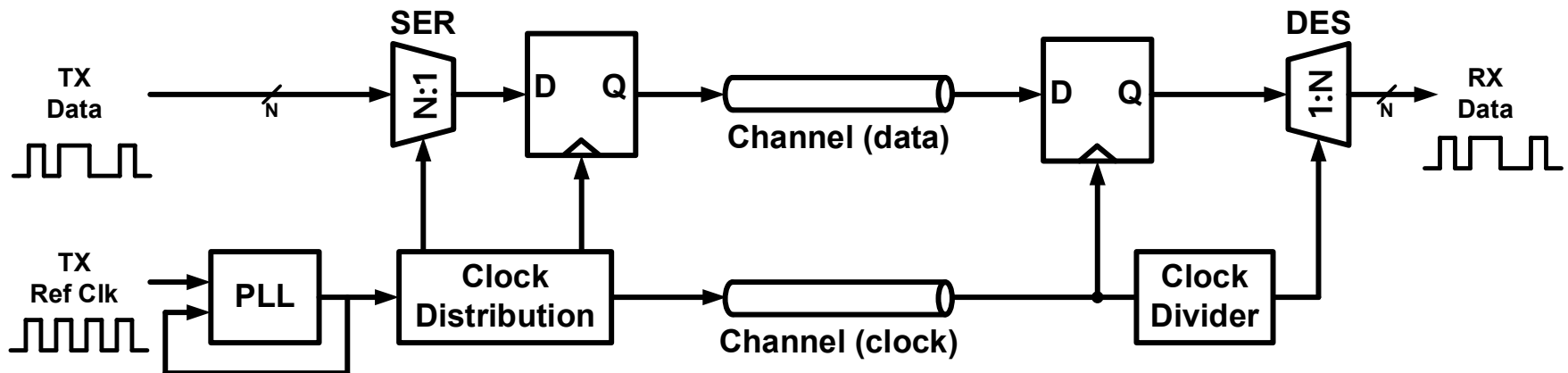
Architecture Evolutions (1)

- Transmission of binary data → Originally a pure digital system
- TX and RX are simply implemented by using a flip-flop
- Then, what is the limitation of this architecture?
 - Needs to transmit massive data



Architecture Evolutions (2)

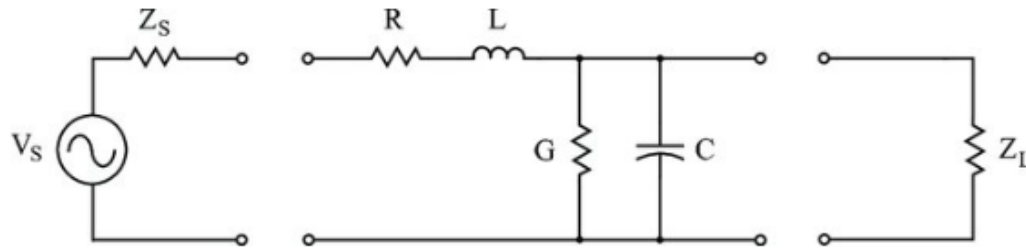
- Serializer, deserializer, and clocking circuits are added
 - Ser/Des circuits decouple the core speed and the interface speed
 - Clocking circuits are needed (PLL, clock distribution, divider...)



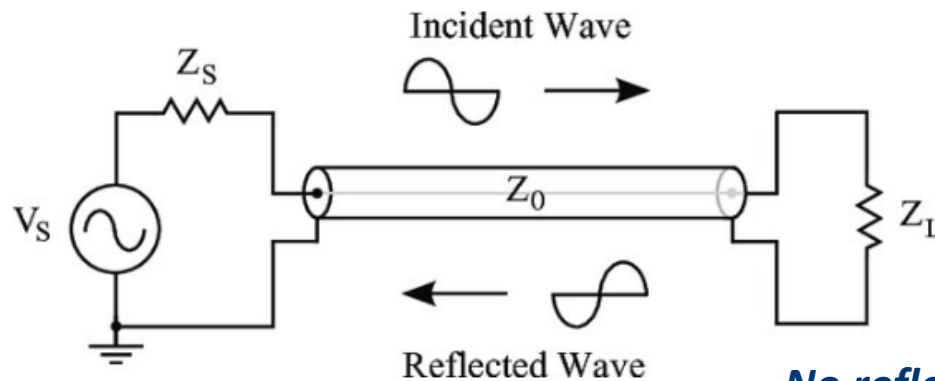
Architecture Evolutions (3)

- Channel is not a simple wire at high frequency
- For high-frequency signal propagation, we treat the channel as a **transmission line**

→ Reflection issue !



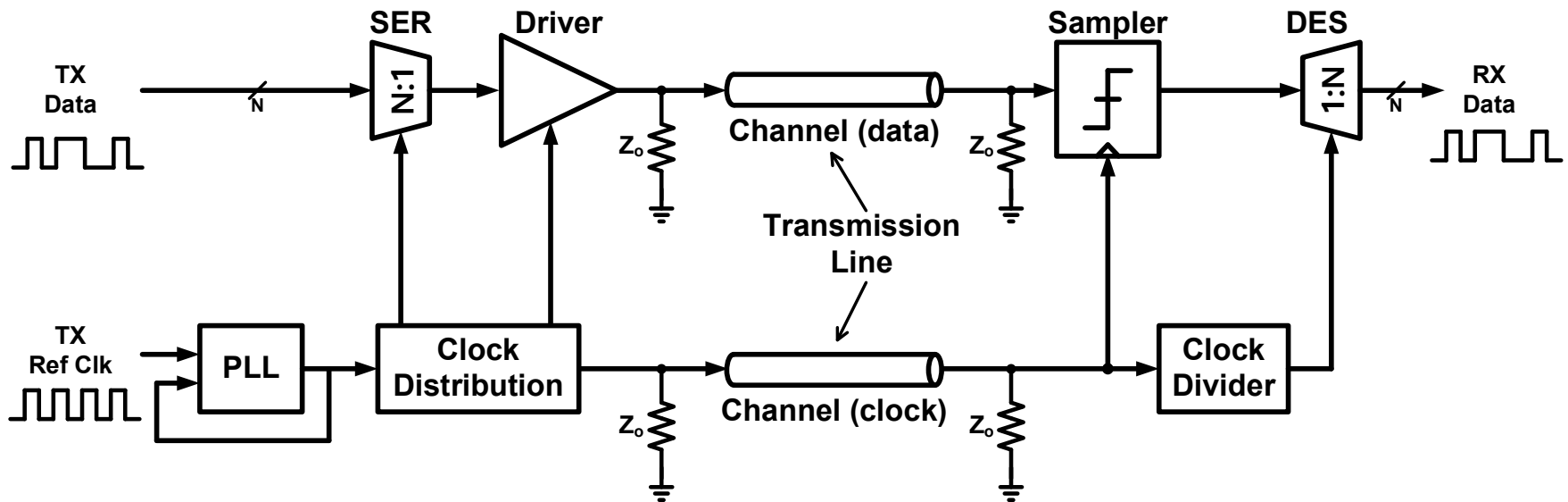
RLGC Transmission Line Model



No reflection if $Z_0 = Z_L$

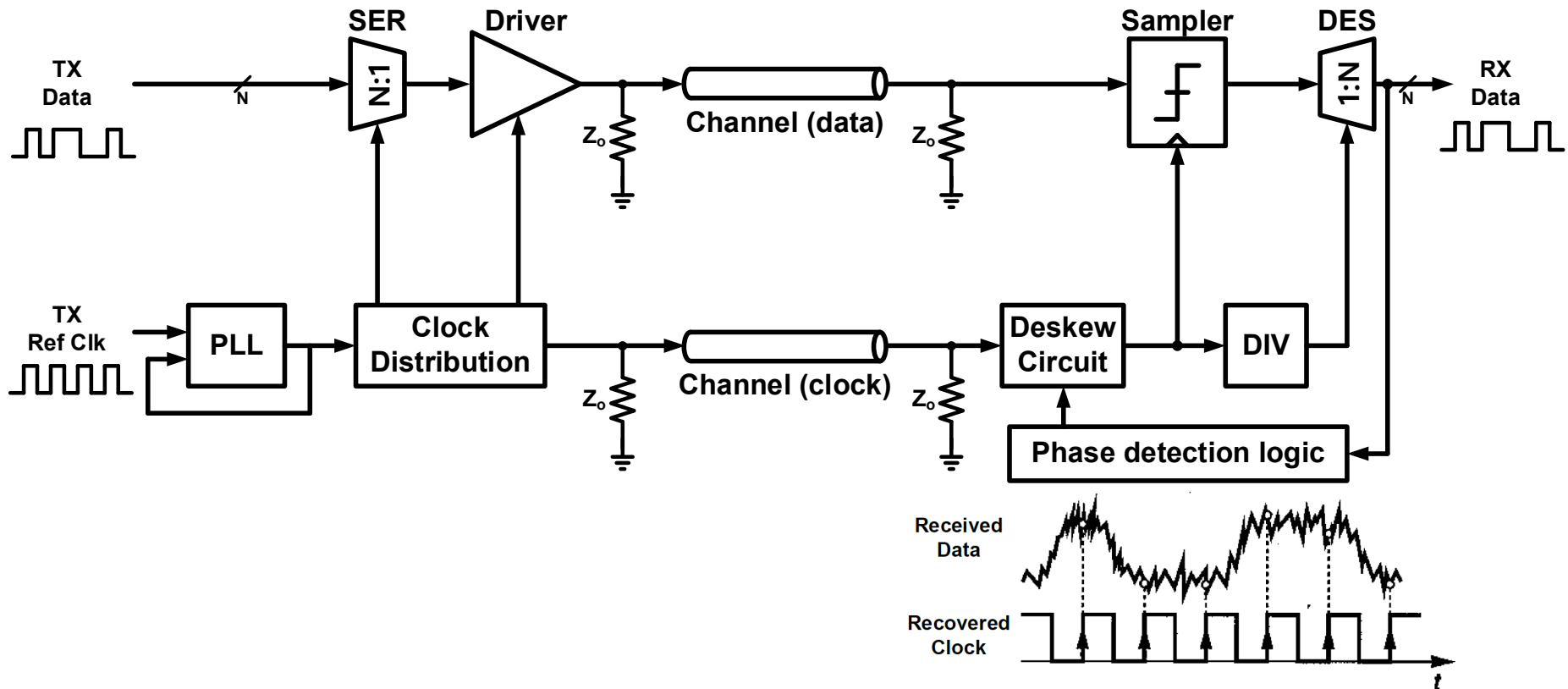
Architecture Evolutions (3)

- Channel is not a simple wire at high frequency
- For high-frequency signal propagation, we treat the channel as a transmission line
- Impedance matching: **termination resistors** are added
- **Driver, buffers, and sampler** are needed for small swing signaling



Architecture Evolutions (4)

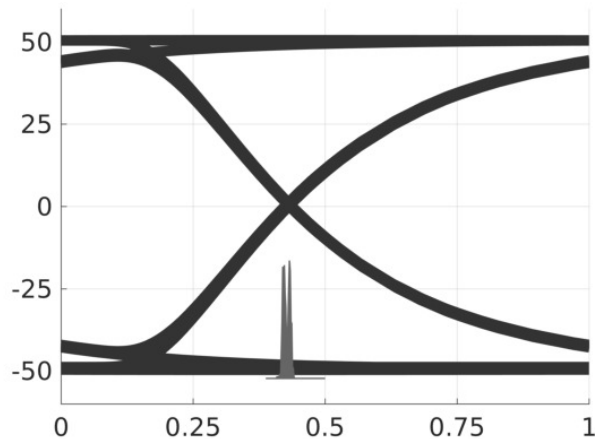
- Timing skew between data path and clock path reduces sampling margin and limits the data rate
 - De-skewing is required
 - **Phase detection logic and de-skew circuits** are added



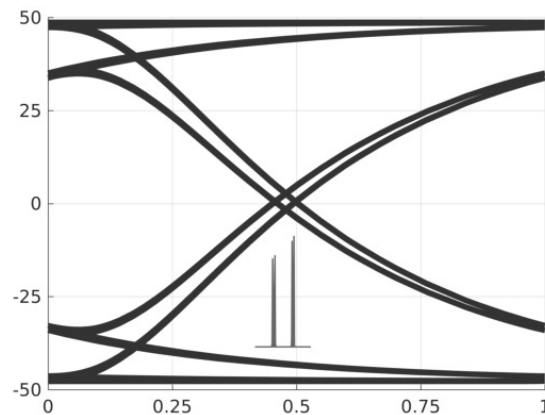
Architecture Evolutions (5)

- **Intersymbol interference (ISI)**

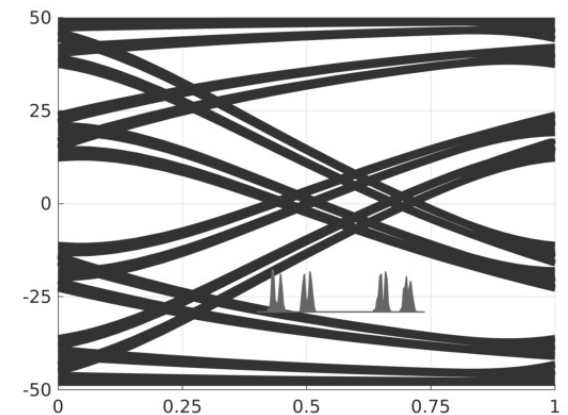
- Previous symbol does not settle completely, then it affects the current symbol
- Data-dependent jitter



Small ISI



ISI spanning over 1 bit

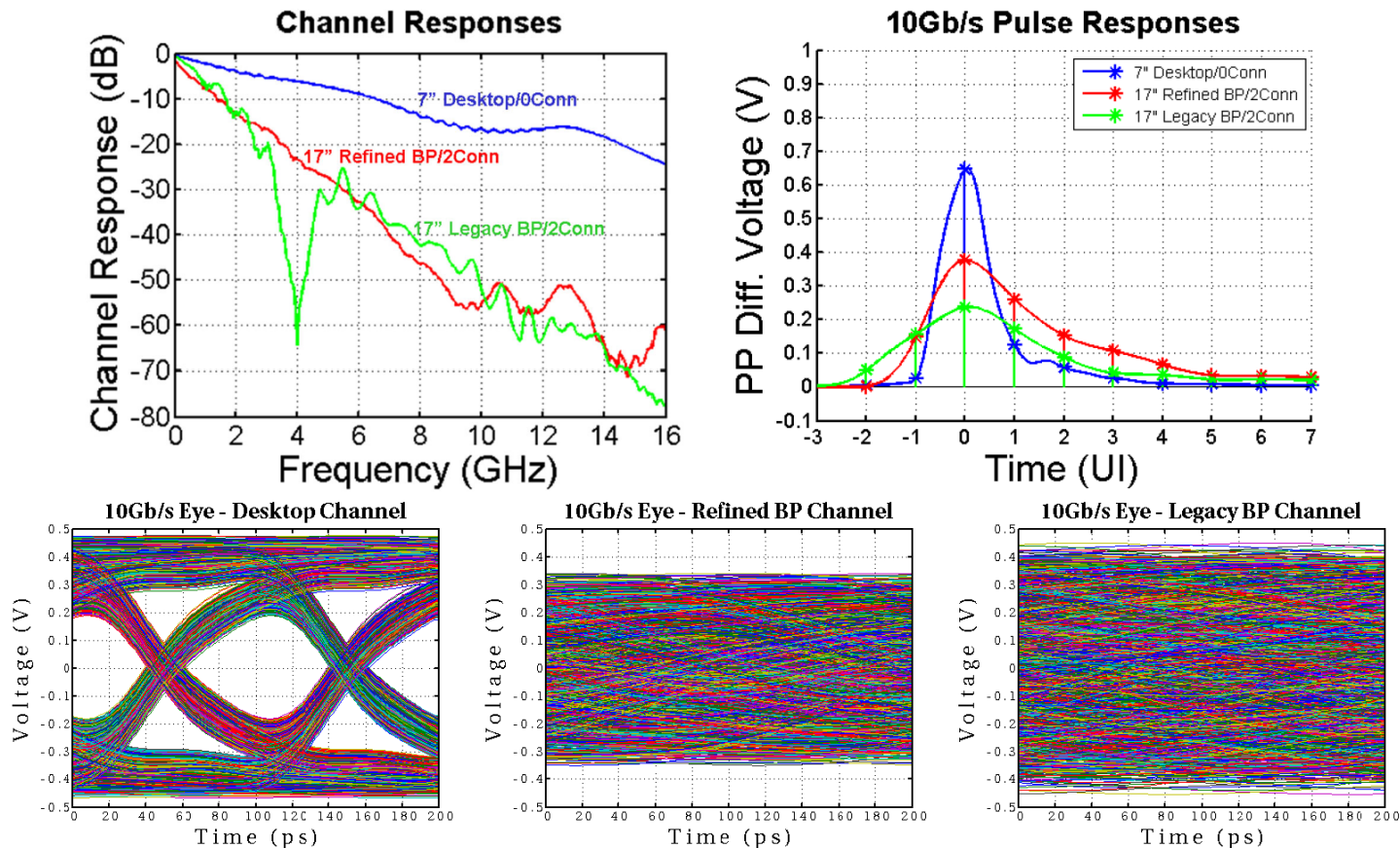


ISI spanning over 2 bits

Architecture Evolutions (5)

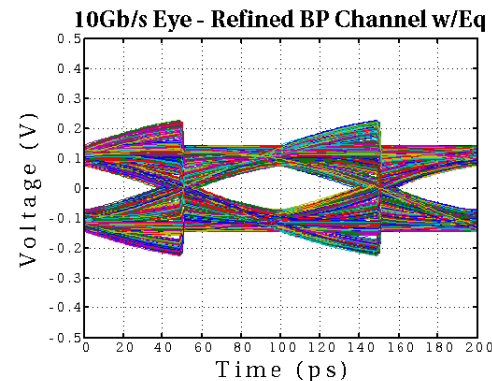
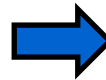
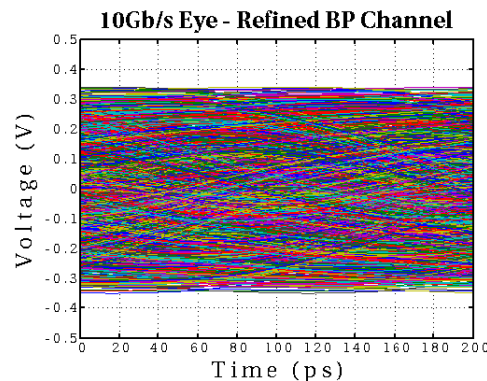
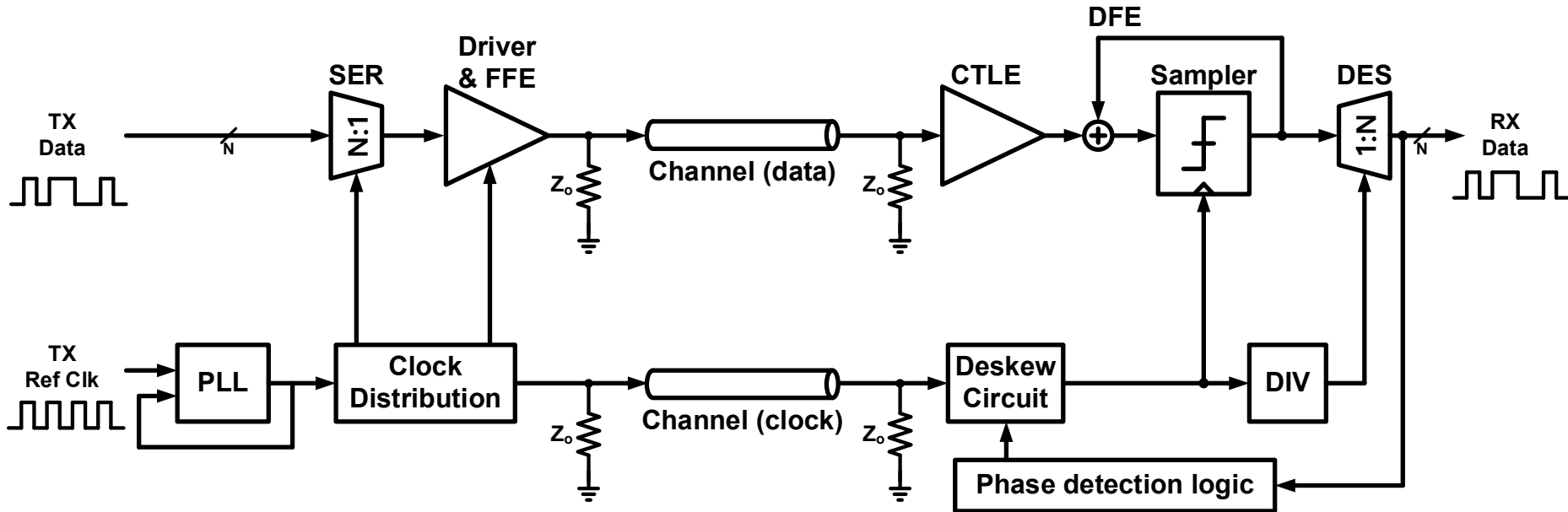
- Intersymbol interference (ISI)

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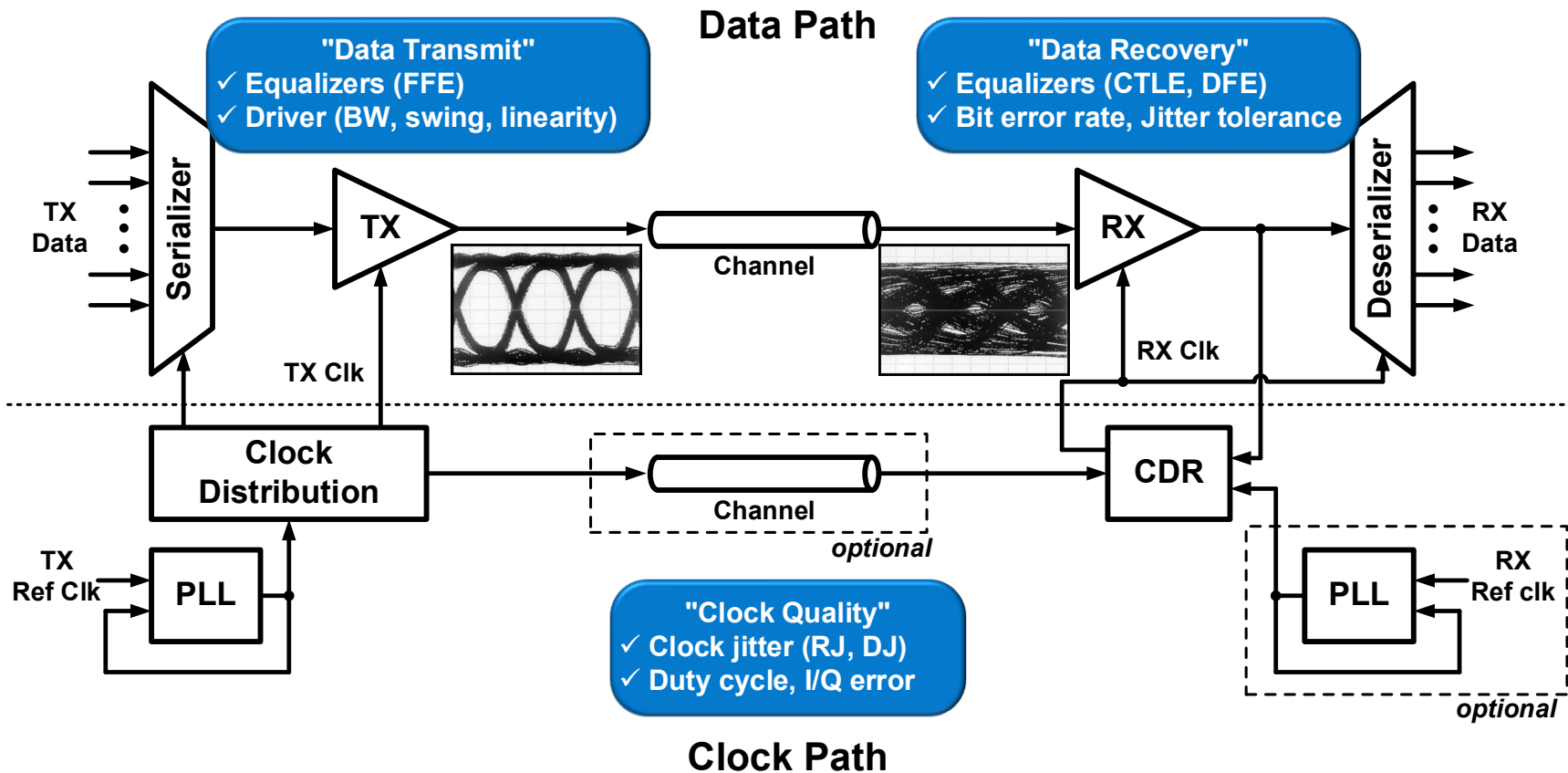
Architecture Evolutions (5)

- Needs to compensate the channel loss
- Equalizer circuits** are added: FFE, CTLE, and DFE



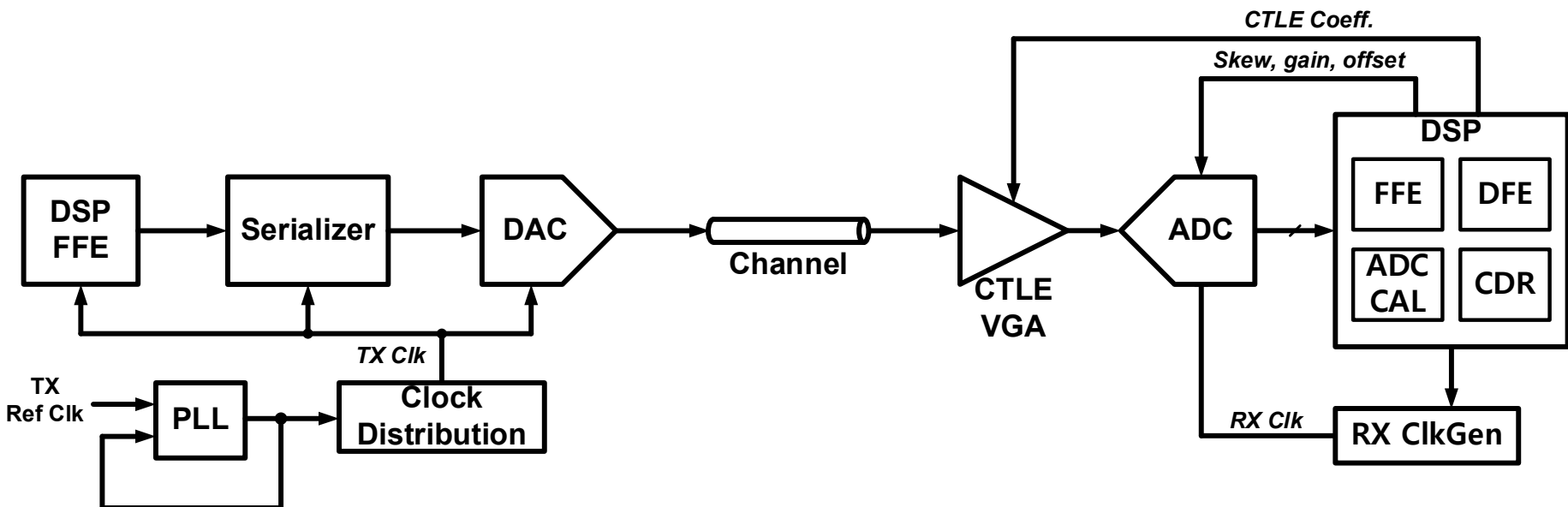
Mixed-Signal Architecture

- High-speed serial link consists of mixed-signal circuits
 - Data path: Driver, Serializer, Equalizer, Sampler, Deserializer
 - Clock path: Phase-locked loop (PLL), Clock and data recovery (CDR)



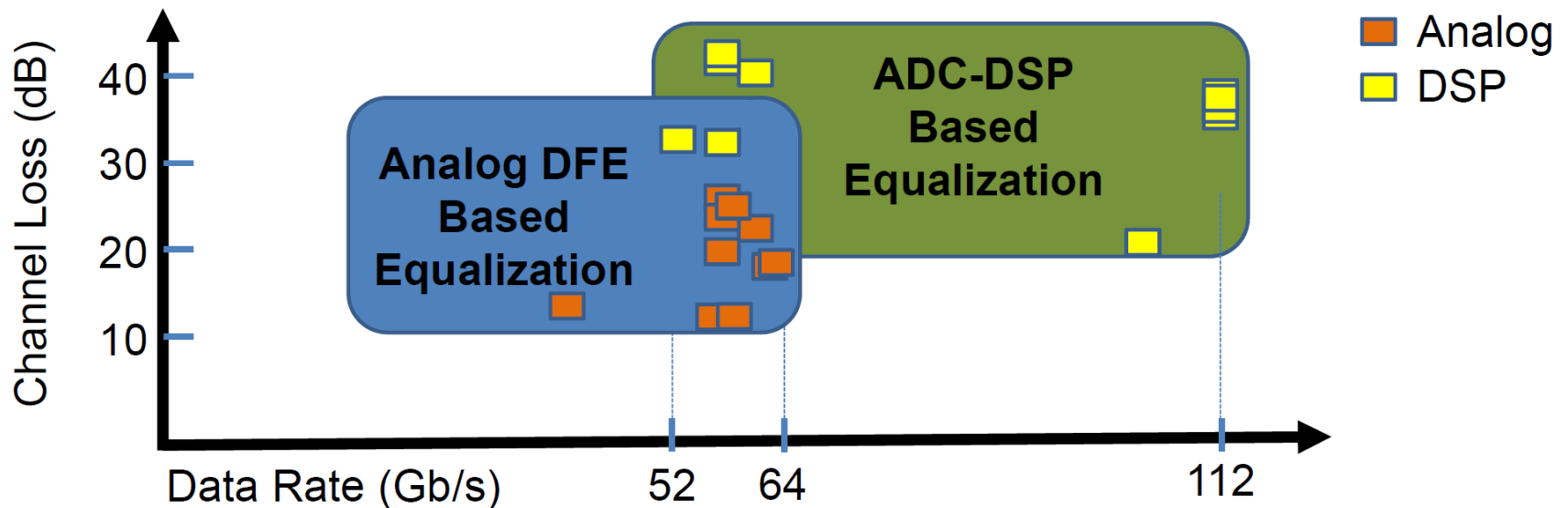
ADC-Based Architecture

- High-speed serial link consists of mixed-signal circuits
 - Front-end consists of DAC and ADC to digitize the analog signal
 - Equalization is done by digital signal processing (DSP)
- Pros and cons
 - Takes advantage of **technology scaling** and enables **advanced equalization** for compensating higher loss ($> 30\text{dB}$)
 - **Large power consumption** of high-speed, high-resolution ADC and **long latency** CDR



ADC-Based Architecture

- ADC-based architecture is dominant in >100Gb/s serial links
- Powerful equalization: channel loss > 30dB
 - Capable of long reach application

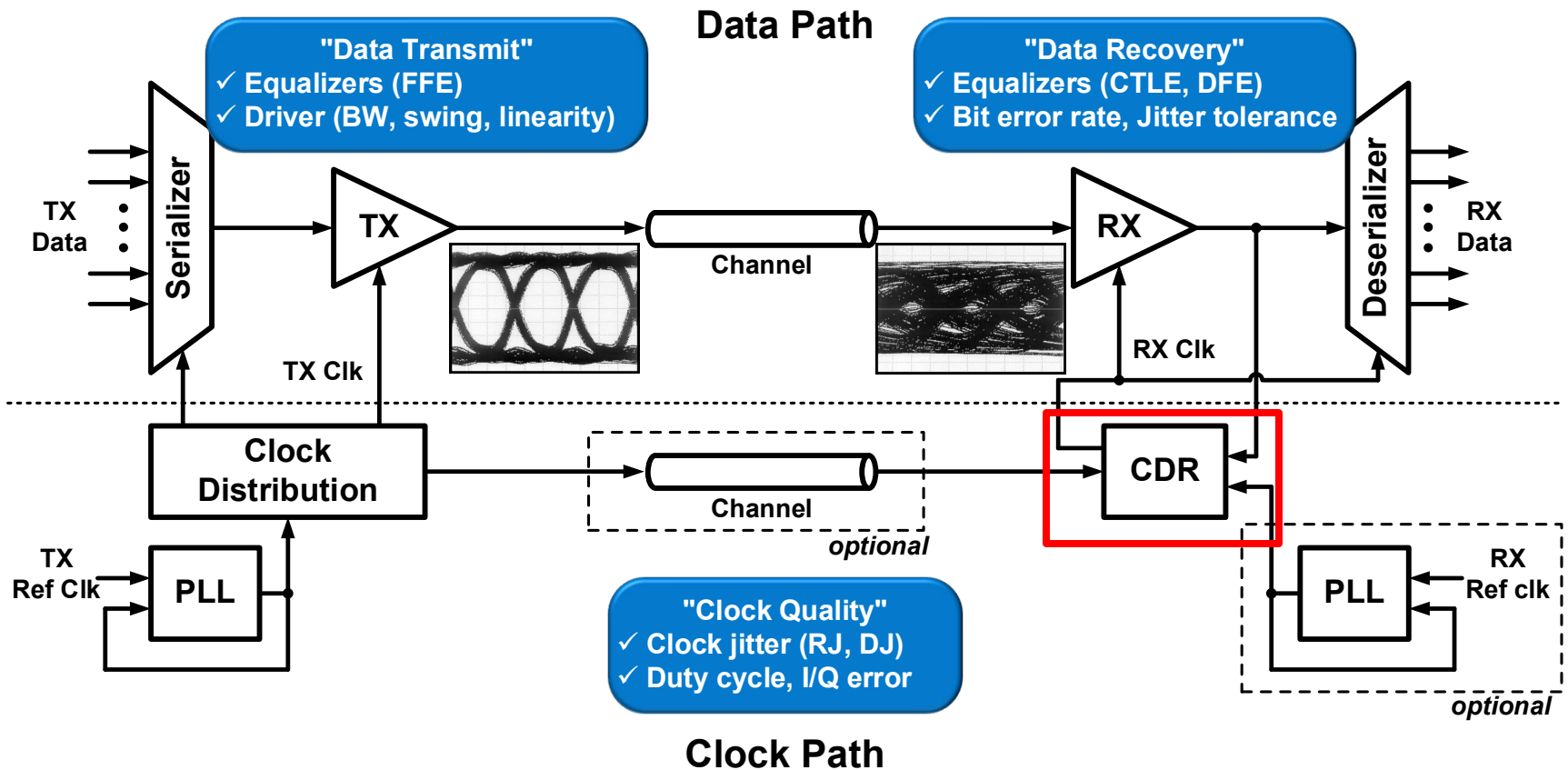


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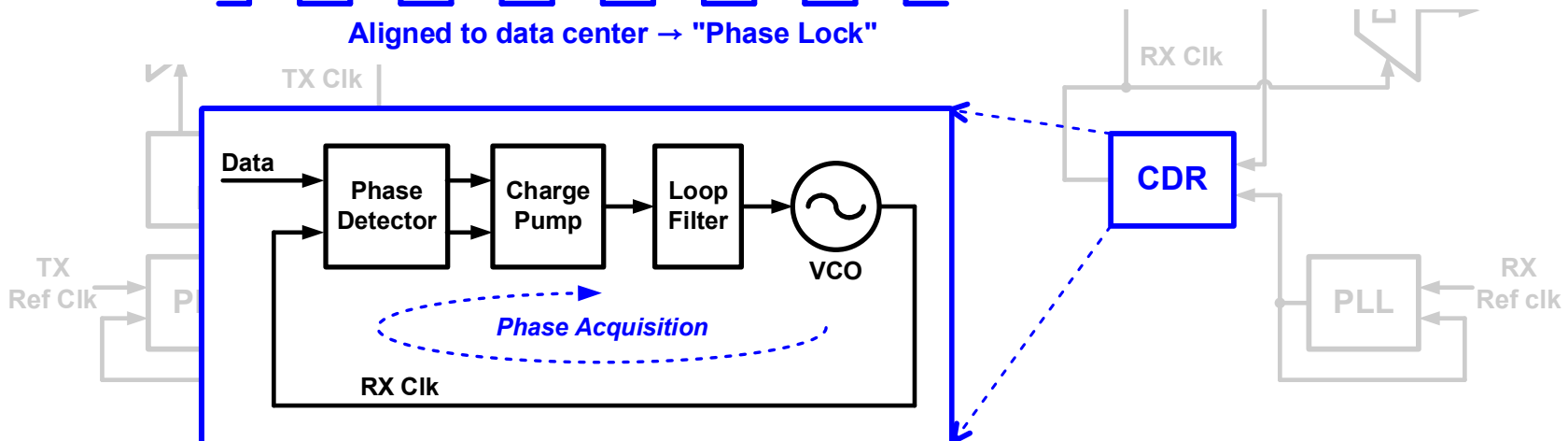
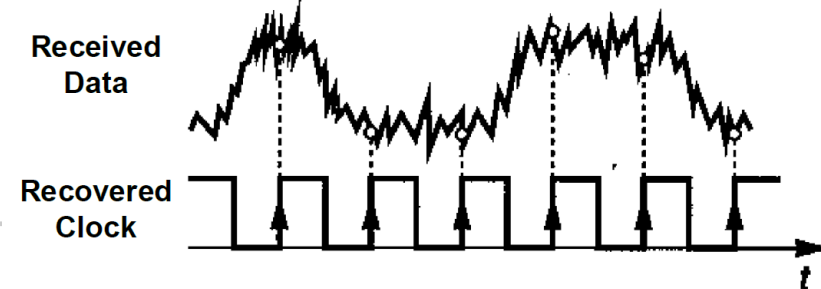
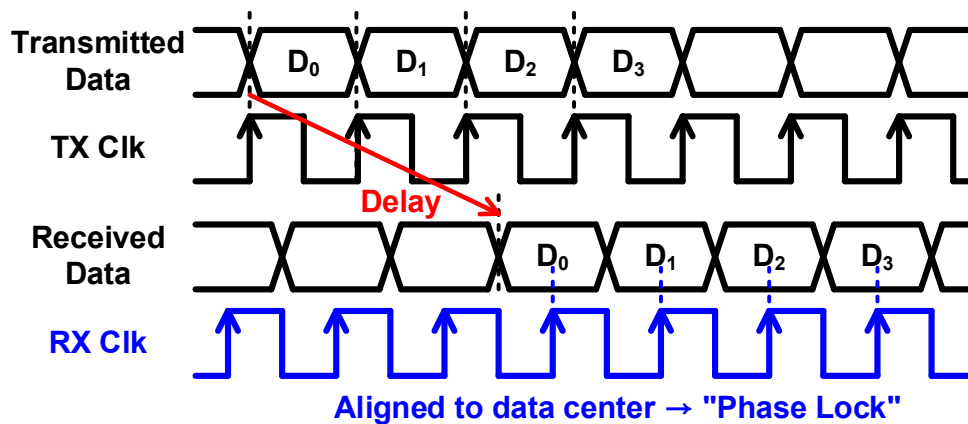
CDR in High-Speed Serial Link

- High-speed serial link consists of mixed-signal circuits
 - Data path: Driver, Serializer, Equalizer, Sampler, Deserializer
 - Clock path: Phase-locked loop (PLL), **Clock and data recovery (CDR)**



Basics of Clock and Data Recovery

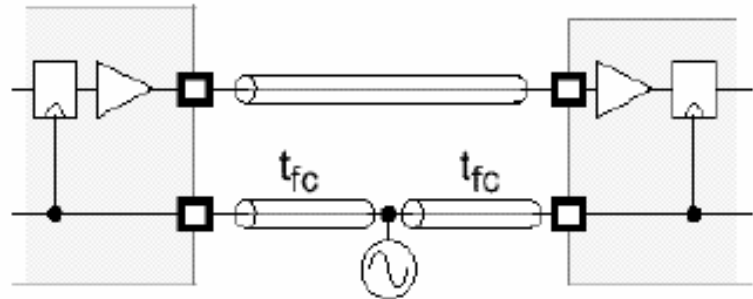
- Clock and Data Recovery (CDR)
 - A CDR circuit produces the clocks to sample incoming data
 - **Phase acquisition** is required to secure sampling margin



Clocking Terminology

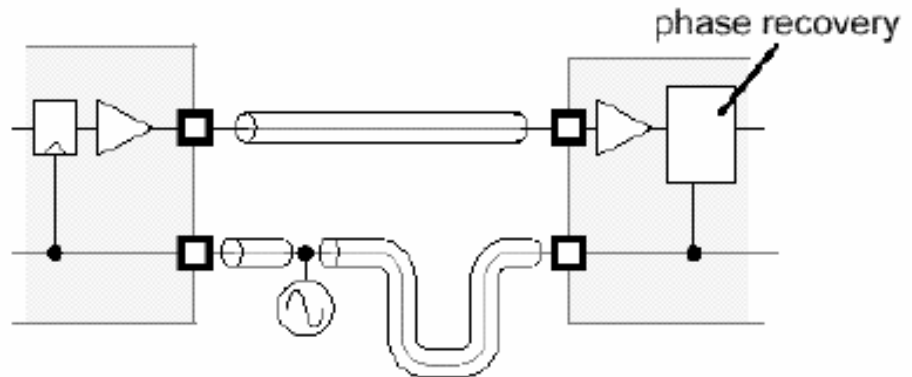
- **Synchronous**

- Every chip gets same frequency and phase
- Used in low-speed busses



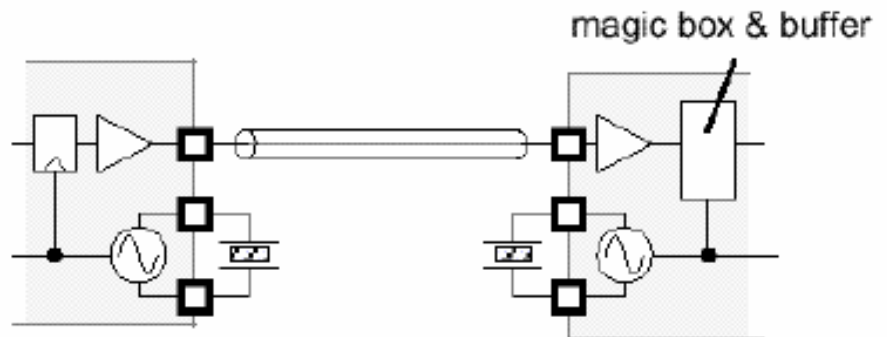
- **Mesochronous**

- Same frequency, but unknown phase
- Requires phase recovery circuit



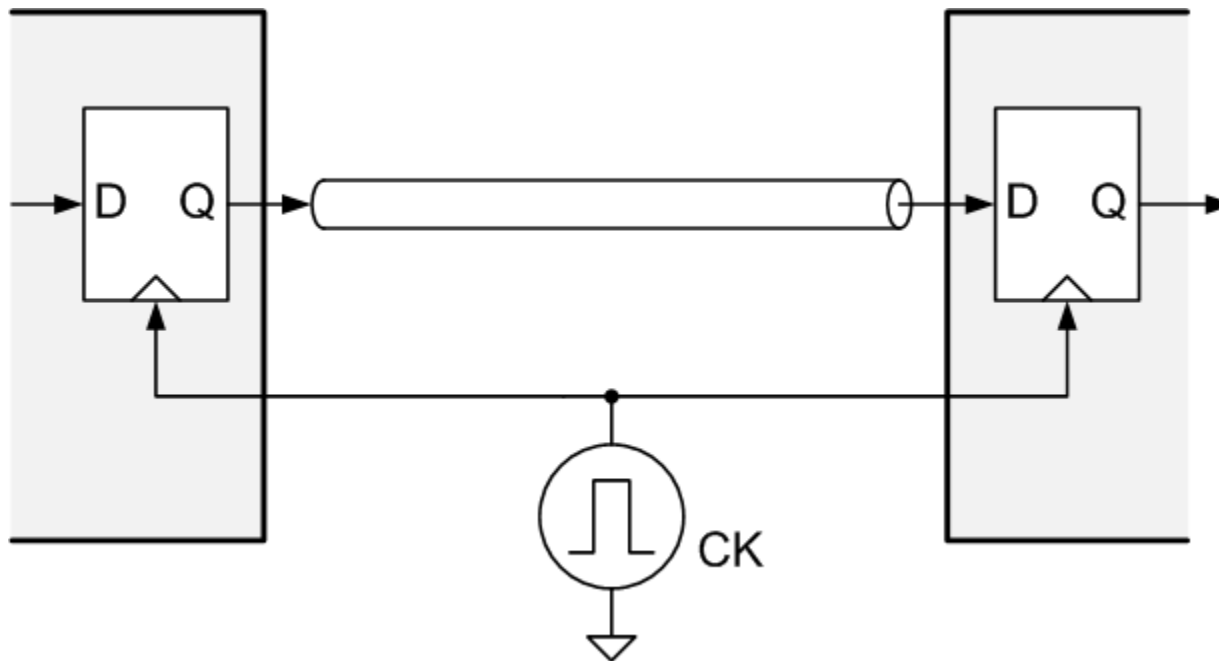
- **Plesiochronous**

- Almost the same frequency, resulting in slowly drifting phase
- Requires CDR
- Widely used in high-speed links



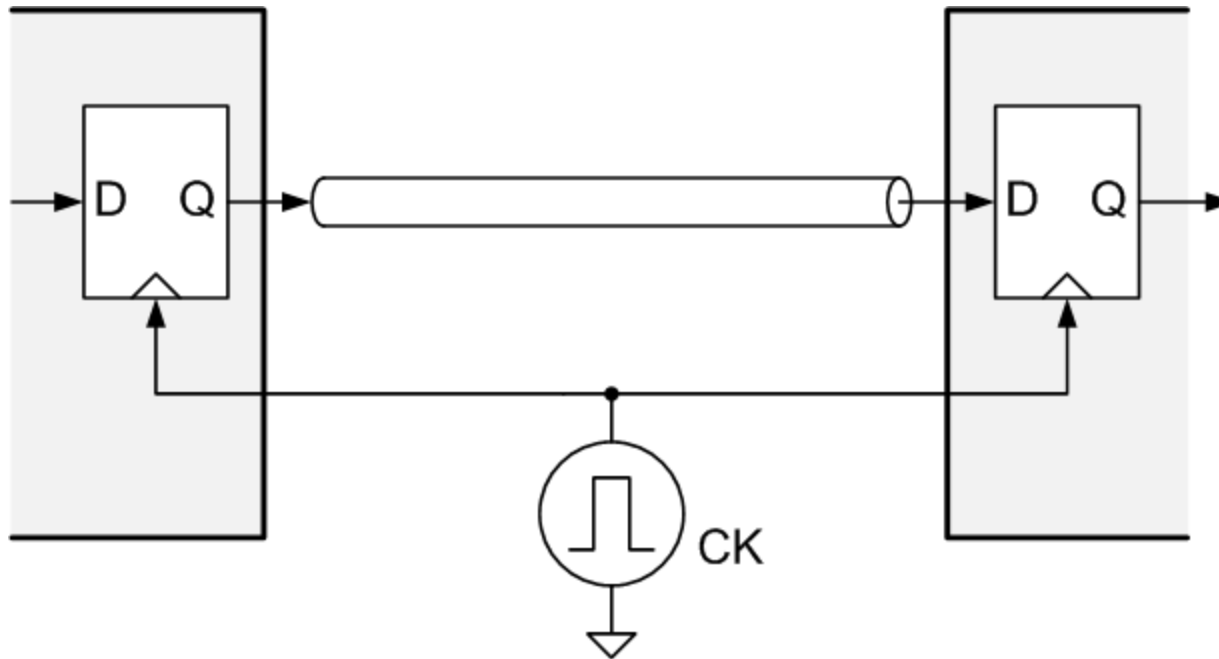
Synchronous Data Transmission

- Both transmitter and receiver are in the same clock domain
- Synchronous system by design (no active deskew)
- **Requires equal length routes** to chips to minimize clock skew



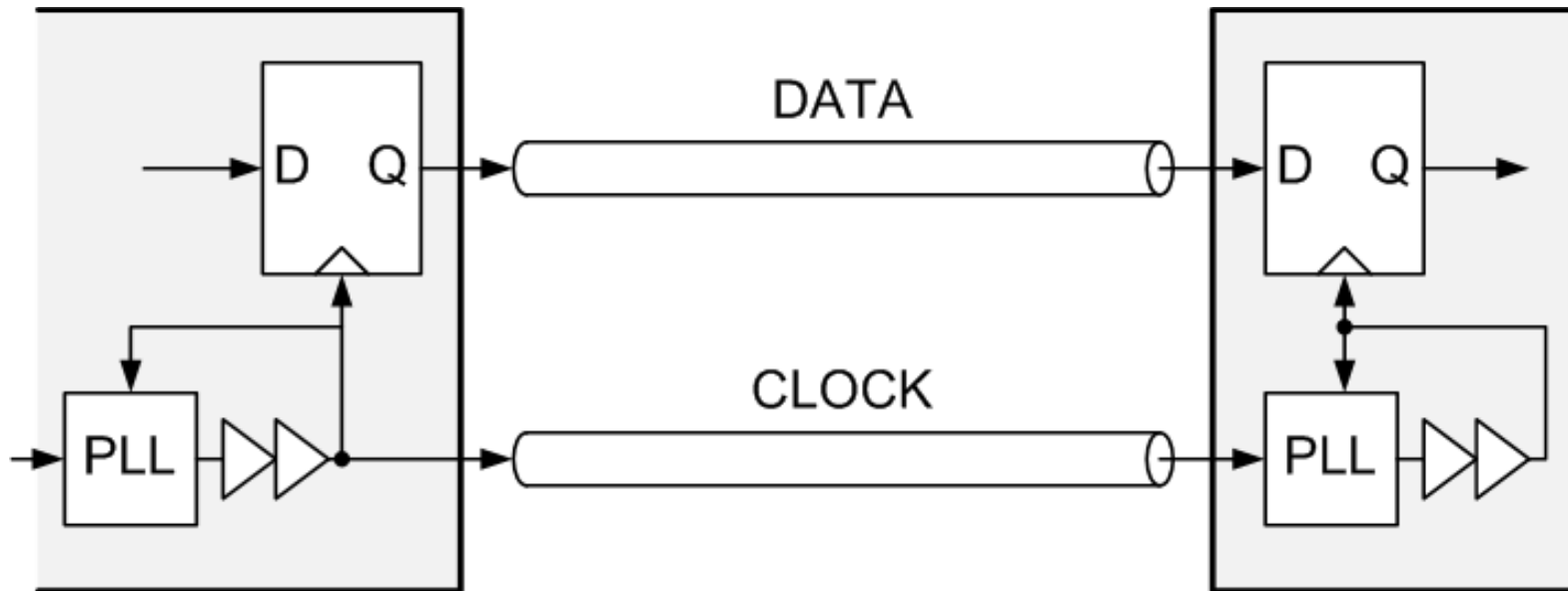
Synchronous Data Transmission

- Both transmitter and receiver are in the same clock domain
- Difficult to control clock skew and propagation delay ☹️
- Sensitive to delay variations in on-chip circuits and board routes ☹️
- What is the maximum possible data rate?
 - How do we decouple the data rate from the channel latency?



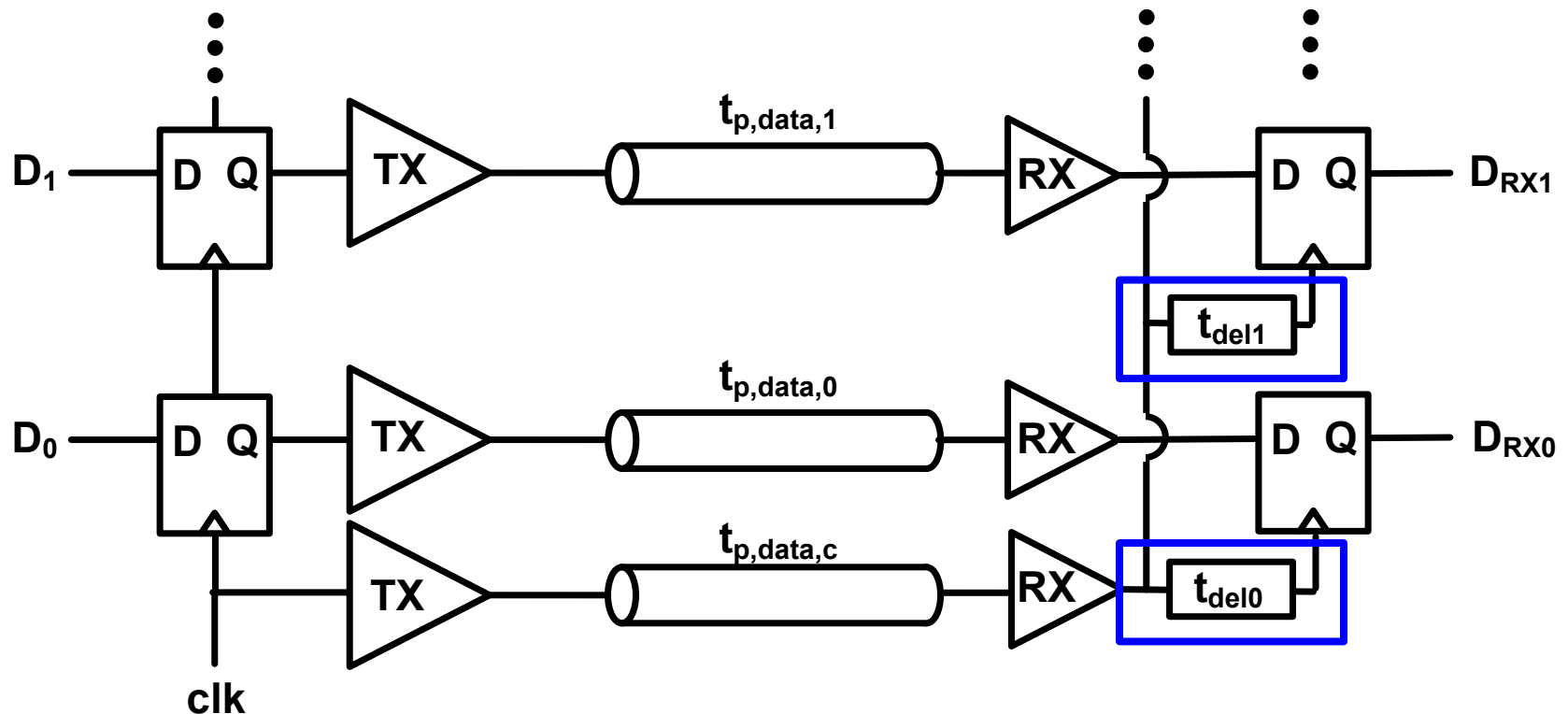
Source Synchronous Link

- Forwarded clocking I/O architecture
 - Same frequency, but unknown phase → Mesochronous system
- Simple de-skewing circuit for phase acquisition 😊
- Needs exclusive clock channel 😞



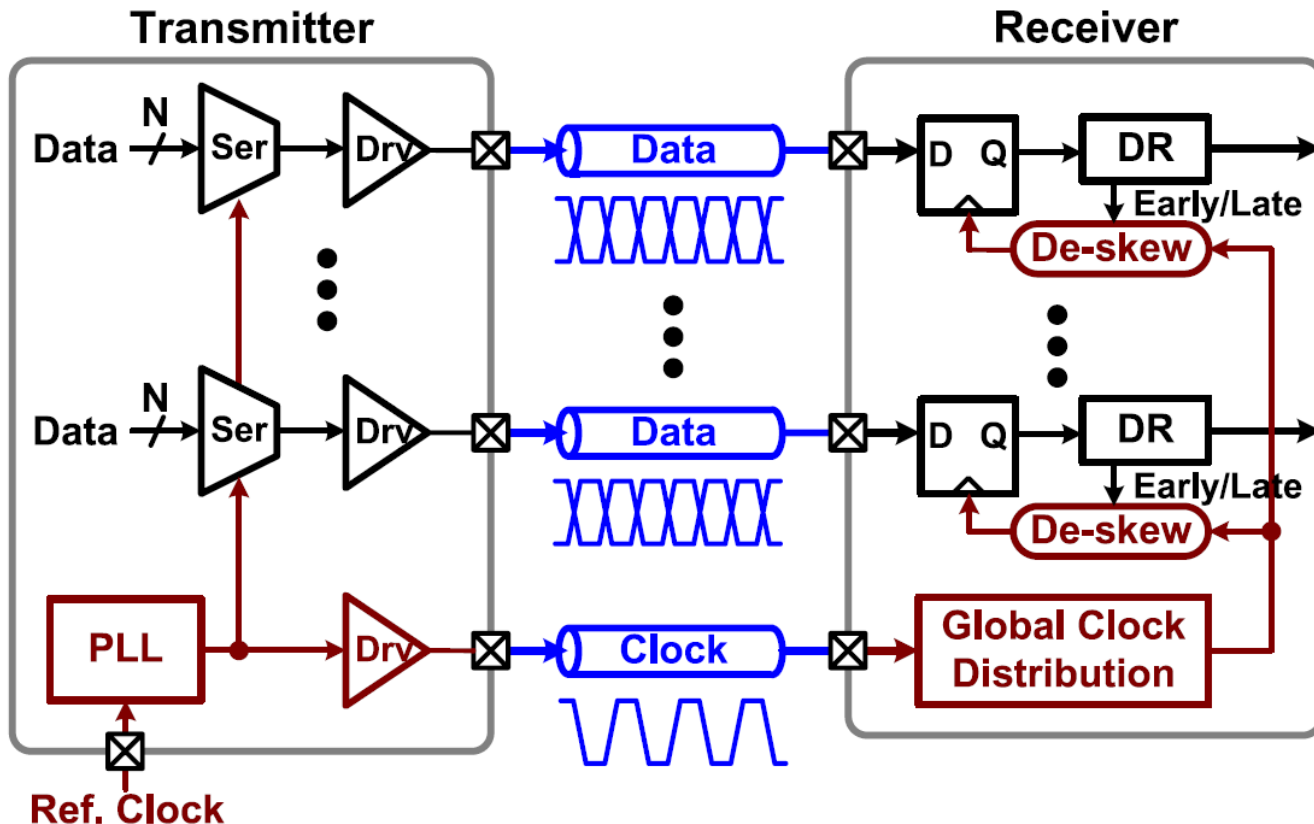
Source Synchronous Link

- Forwarded clocking in multi-channel system
 - Cost of exclusive clock channel can be amortized
 - Skews between the links can be problematic
 - Needs "per-pin" skew compensation

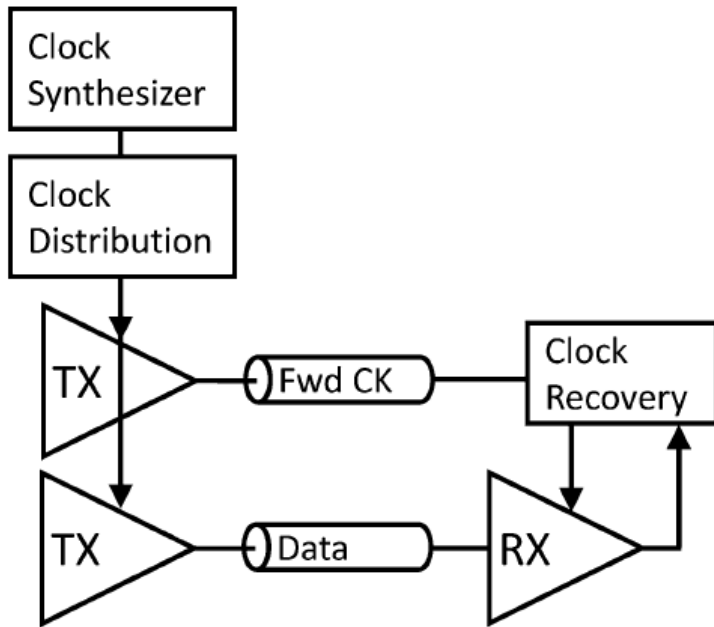


Source Synchronous Link

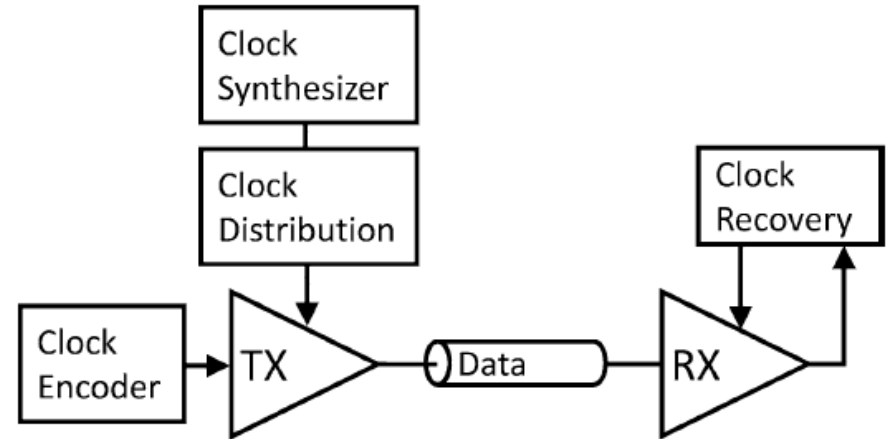
- Forwarded clock receiver can use different **de-skewing circuits**
 - PLL (phase-locked loop), DLL (delay-locked loop), PI (phase interpolator), and ILO (injection-locked loop)



Embedded Clocking



- Forwarded clock architecture
 - Frequency is matched 😊
 - Great matching between clock and data 😊
 - Needs dedicated clock channel 😞



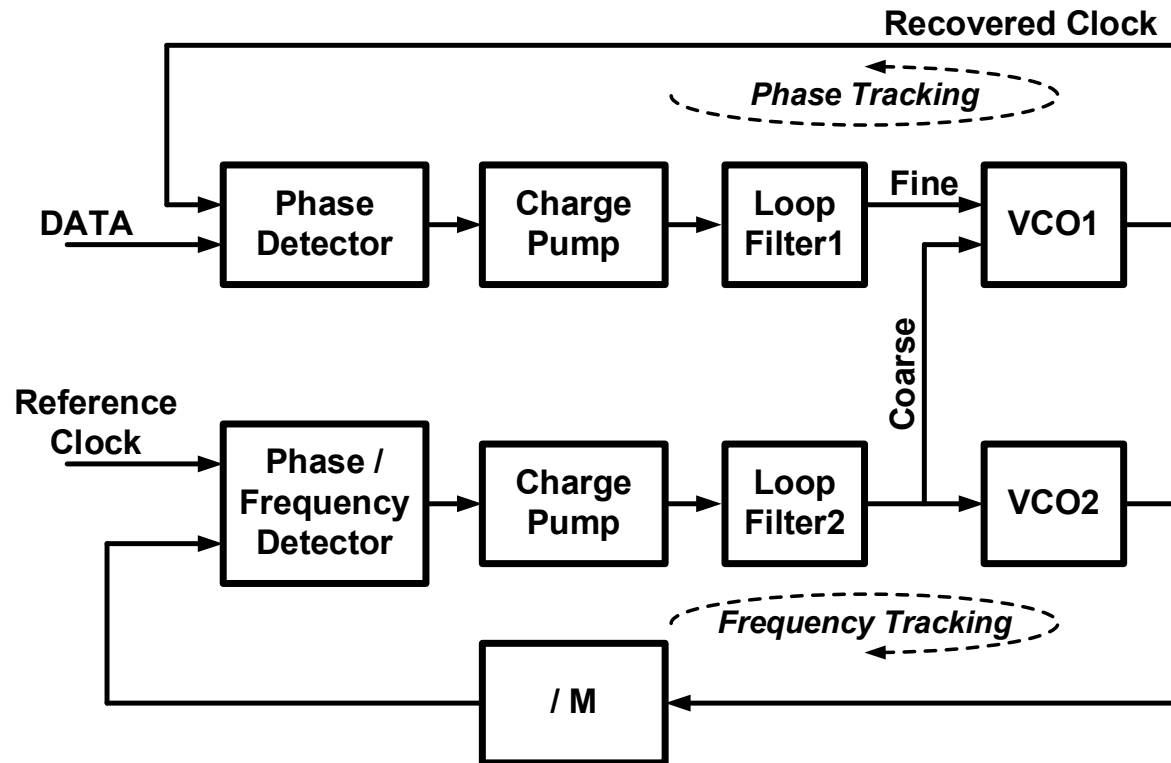
- Embedded clock architecture
 - No need for dedicated clock channel (system cost ↓) 😊
 - Needs to generate clock for data recovery (complicated RX design) 😞

CDR Architectures

- Phase tracking CDR
 - Phase alignment using feedback loop
 - Phase-locked loop (PLL)
 - Delay-locked loop (DLL)
 - Phase interpolator (PI)
 - Injection-locked oscillator (ILO)
- Open-loop CDR
 - Phase alignment without feedback loop
 - Gated oscillator and high-Q bandpass filter
- Oversampling-based CDR
 - No phase alignment
 - Blind oversampling CDR

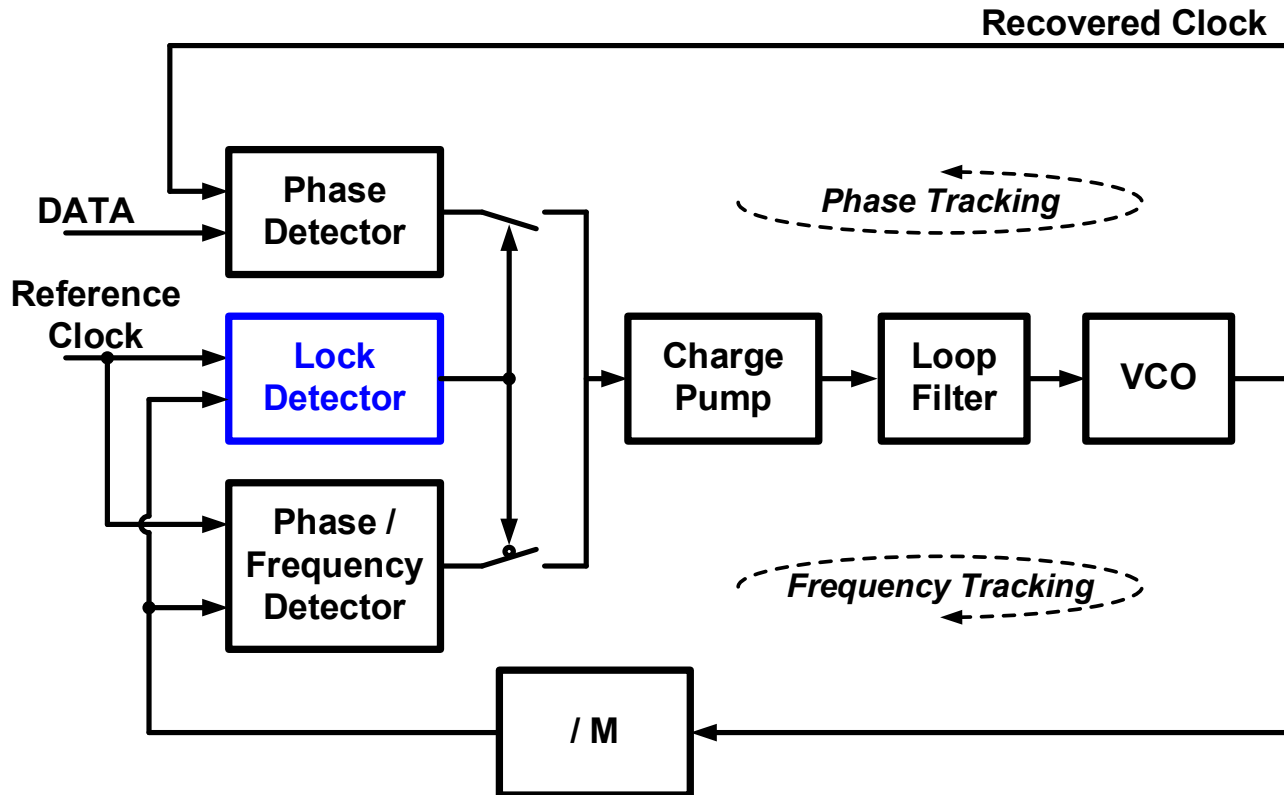
CDR Architectures – Phase Tracking CDR

- **PLL-based CDR with an external reference clock**
 - Two tracking loops for frequency and phase drive coarse and fine control
 - Two VCOs sharing control voltage have mismatch issue and causes large overhead



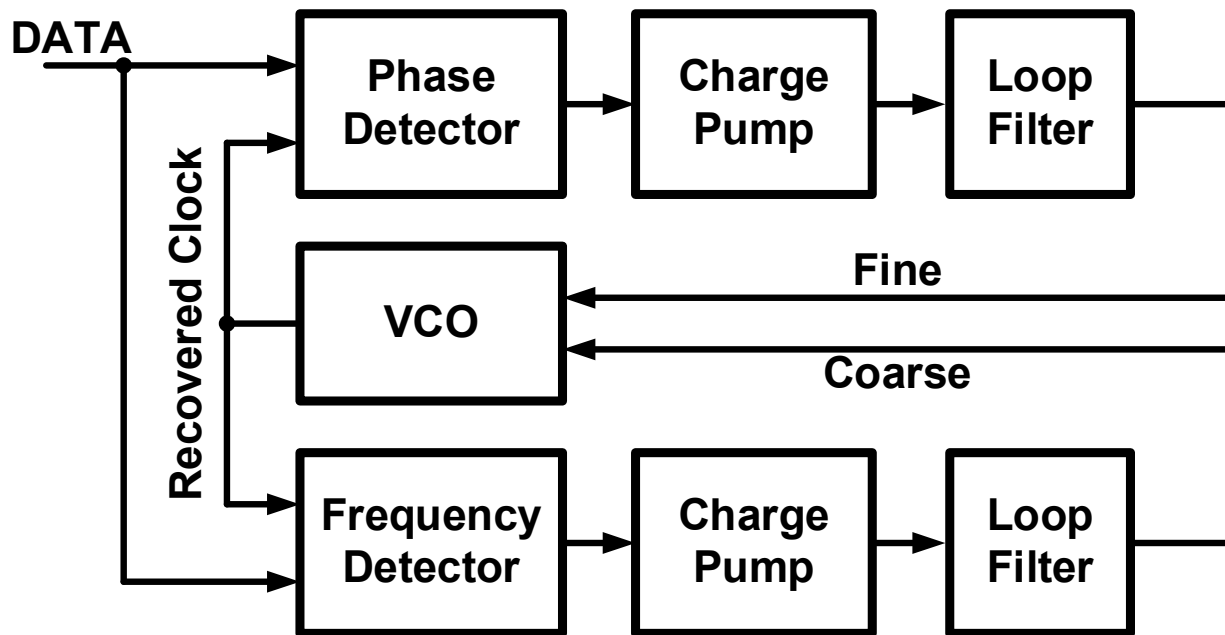
CDR Architectures – Phase Tracking CDR

- PLL-based CDR with a reference clock and a lock detector
 - A lock detector sequentially enable the frequency and phase tracking loop
 - Reduces the hardware overhead



CDR Architectures – Phase Tracking CDR

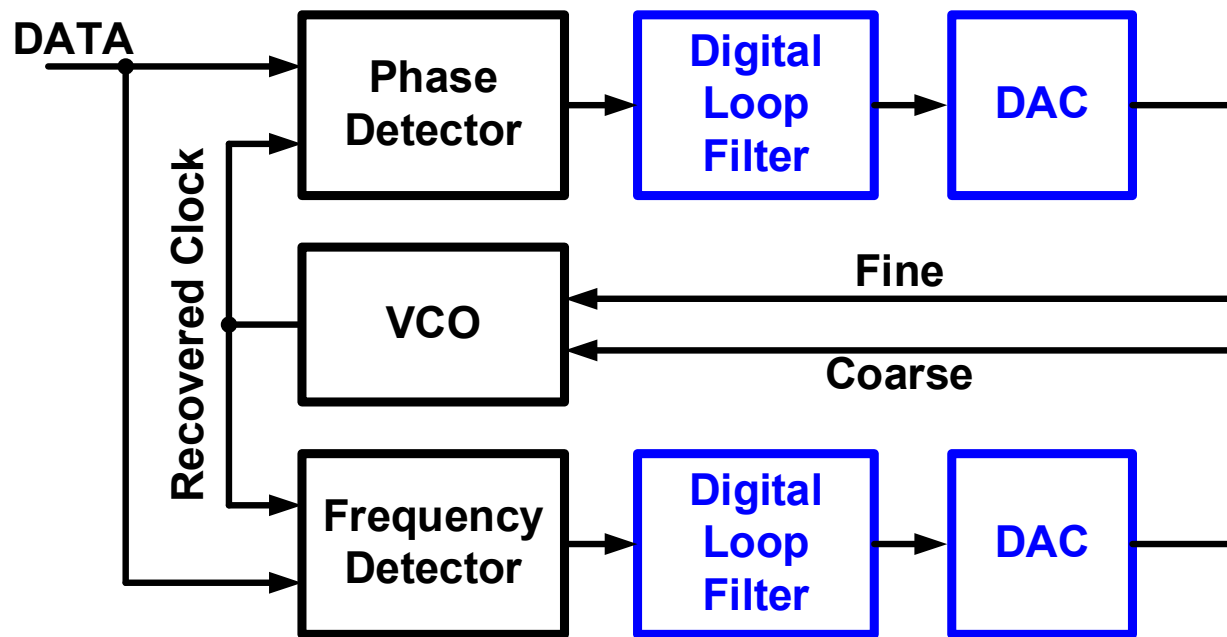
- PLL-based CDR without a reference clock
 - Referenceless CDR
 - Rather than using a reference clock, a frequency detector extracts frequency information from the random data
 - Challenges of designing the frequency detector



CDR Architectures – Phase Tracking CDR

- Digital PLL (DPLL)-based CDR

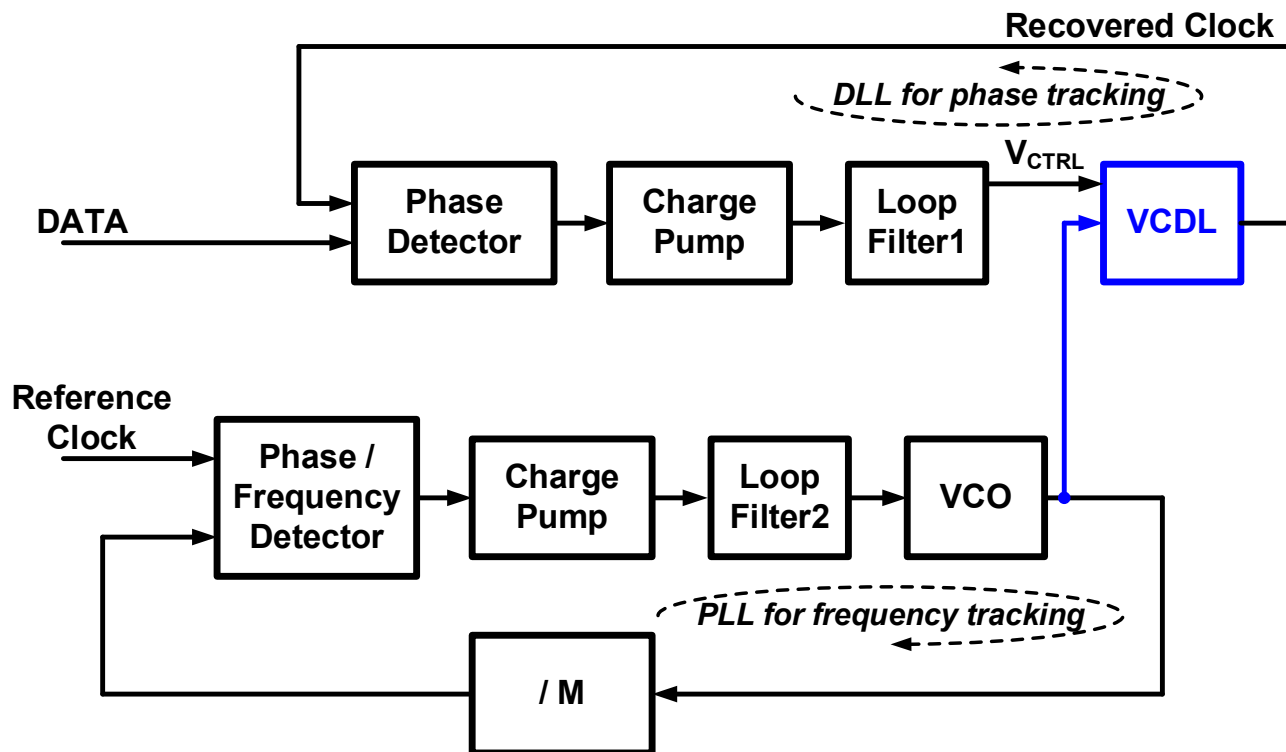
- A digital loop filter can reduce the required layout area
- Reduced PVT variations and leakage issue
- Loop latency issue and quantization noise



CDR Architectures – Phase Tracking CDR

- DLL-based CDR

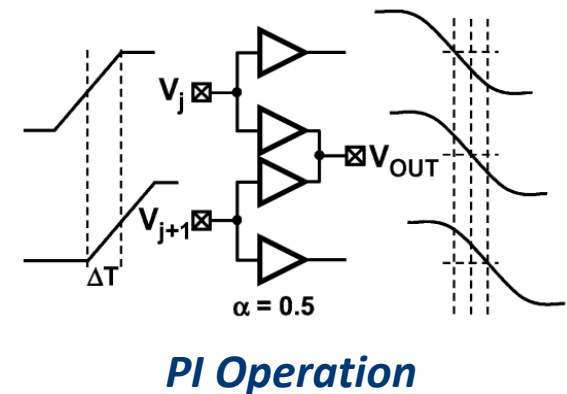
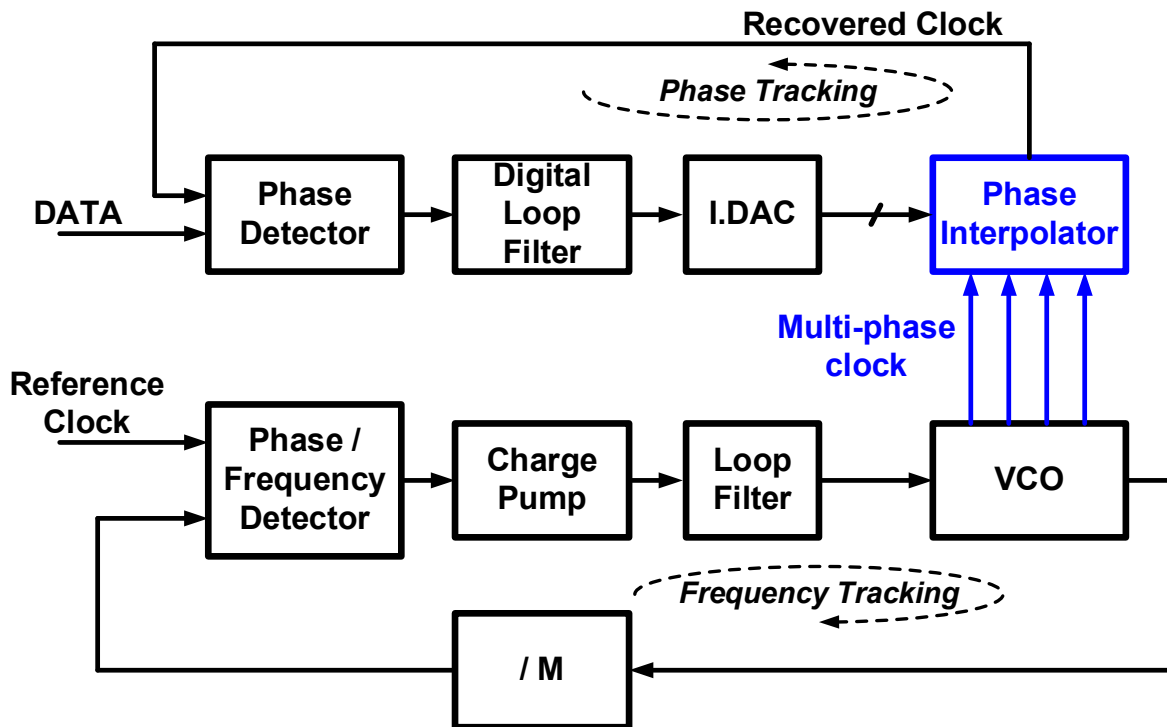
- Avoids drawbacks of multi-VCO mismatches and high power consumption
- No issues of jitter accumulation and loop stability
- Limited phase capturing range and no jitter filtering
- Issues of stuck locking and harmonic locking



CDR Architectures – Phase Tracking CDR

- **PI-based CDR**

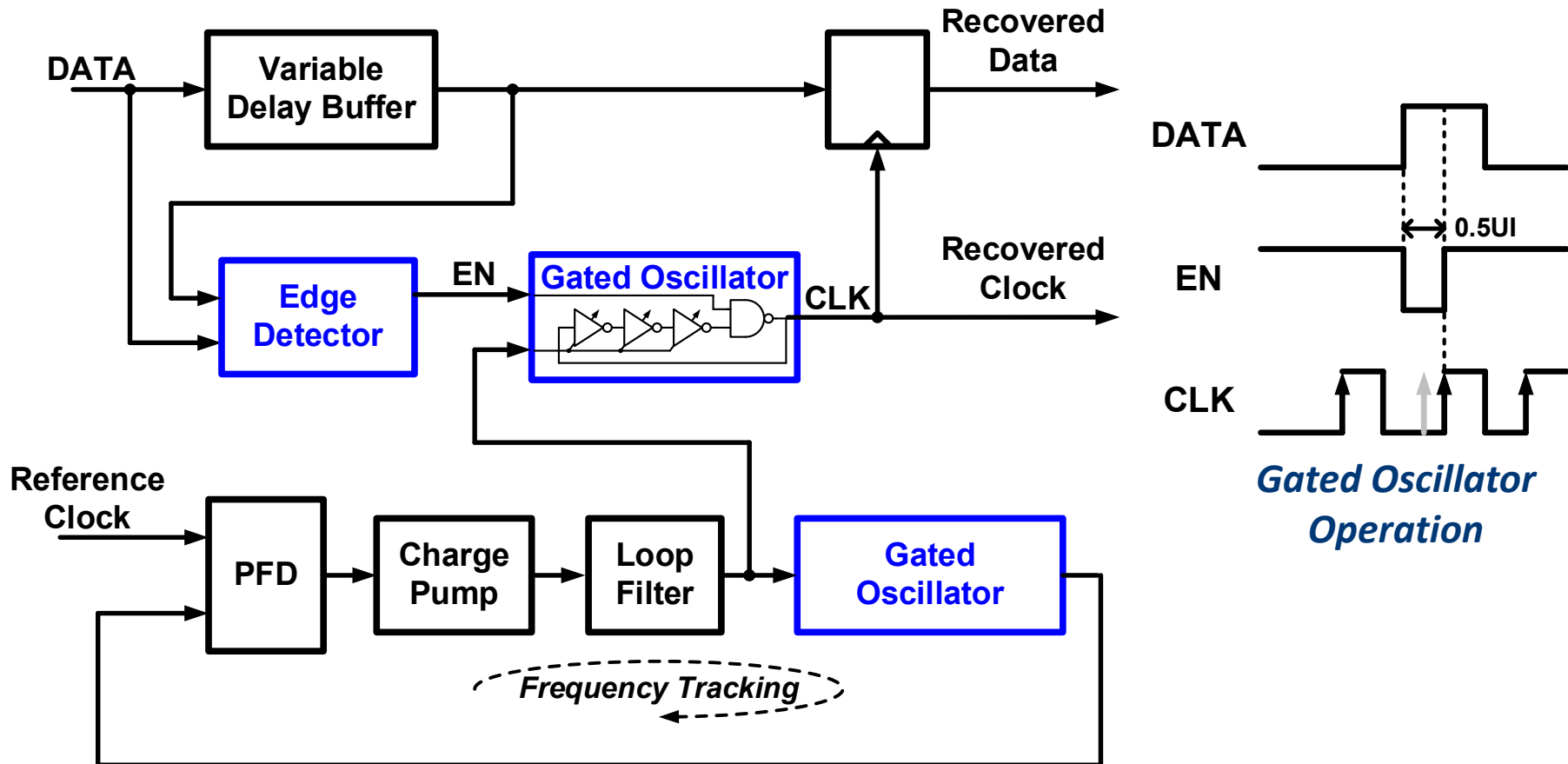
- Stable loop characteristics similar to the DLL-based CDR
- Possible to track a certain allowable frequency offset
- Advantages and disadvantages of digital implementation
- Issues of linearity and multi-phase generation



CDR Architectures – Open-Loop CDR

- **Gated oscillator-based CDR**

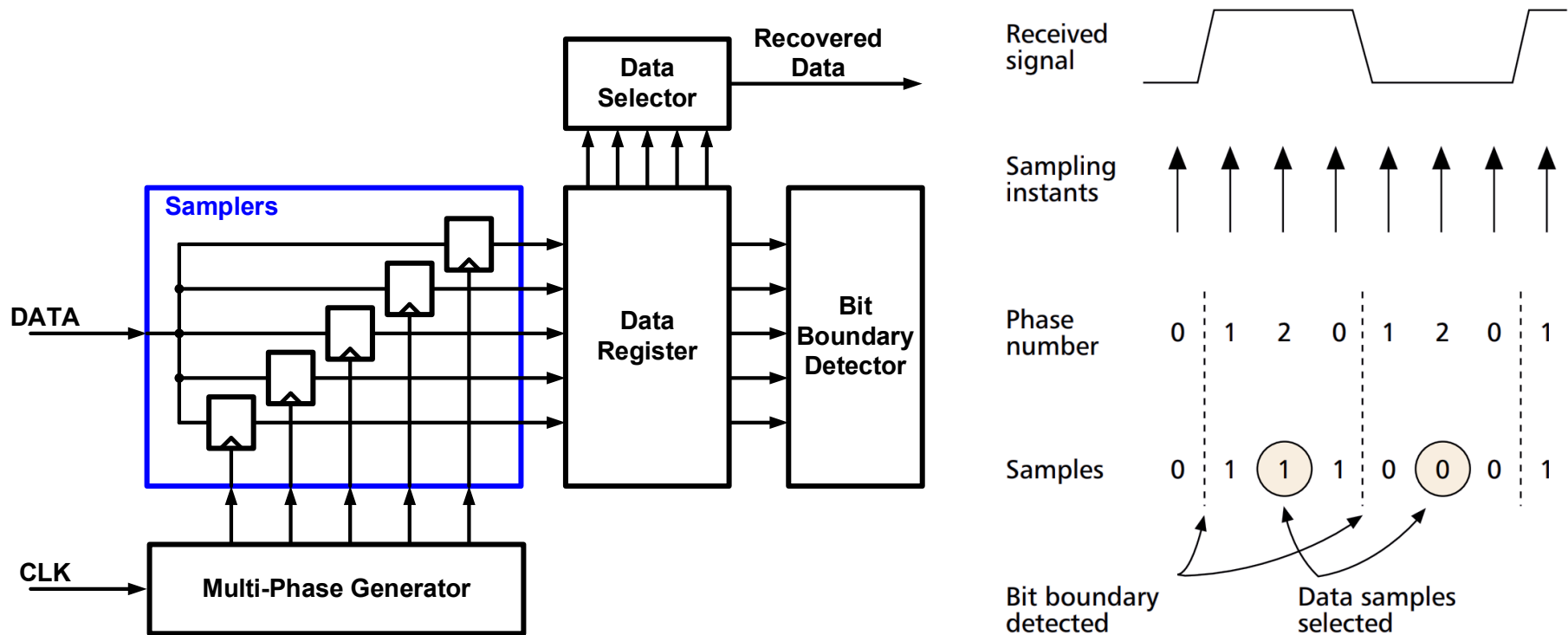
- Fast synchronous clock recovery and data acquisition
- Simple and small design
- No jitter rejection and sensitive to PVT variations



CDR Architectures – Oversampling-based CDR

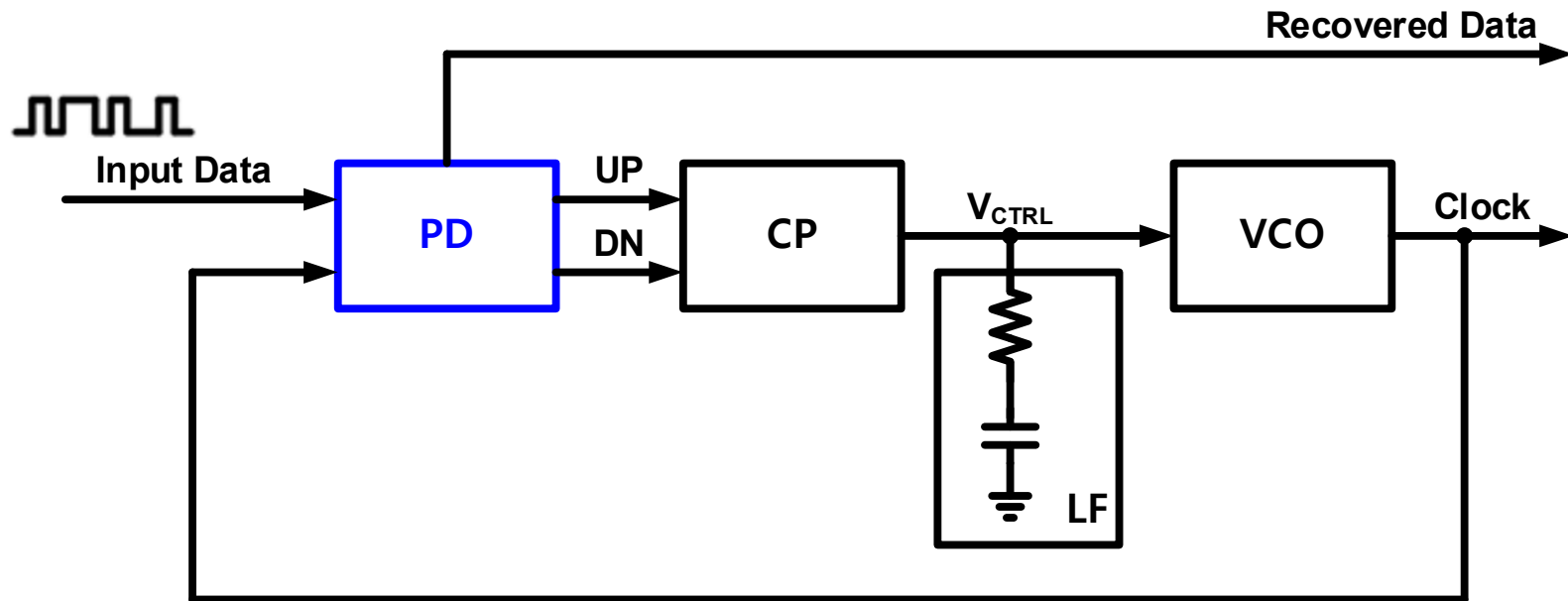
- **Blind oversampling CDR**

- Samples each received data bit at multiple points
- No feedback → fast acquisition and inherent stability
- Needs many sampling circuits, multi-phase clocks, and large data register



CDR Phase Detector

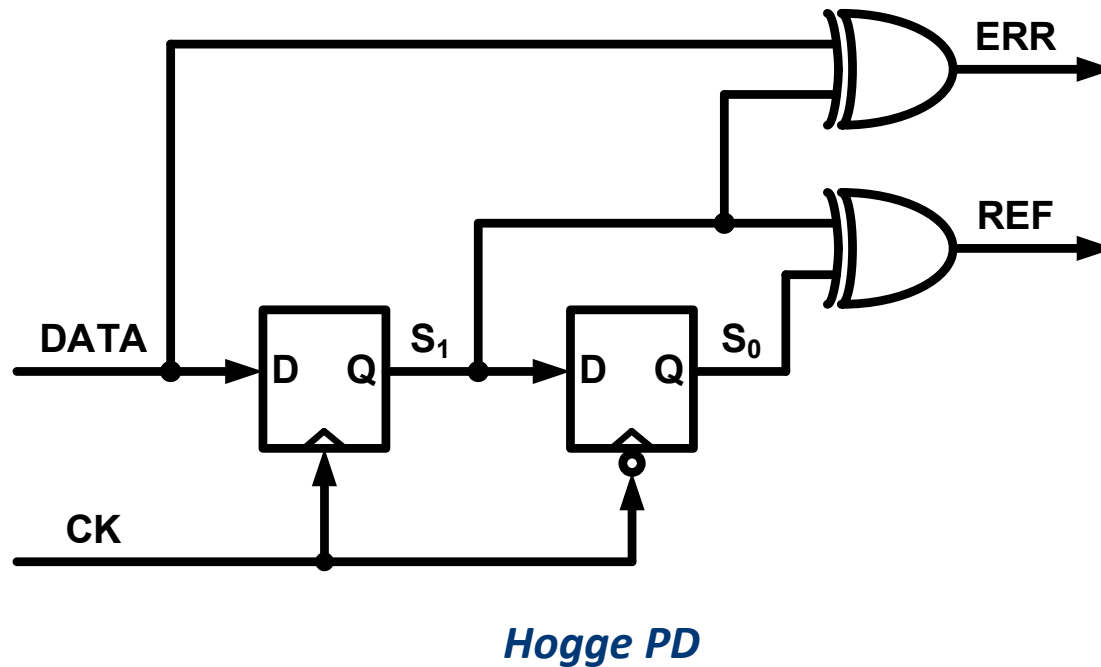
- A primary difference between CDRs and PLLs is that the input data is not periodic but random
- Phase detector should operate properly with missing transition edges



CDR Phase Detectors – Hogge PD

- Hogge phase detector

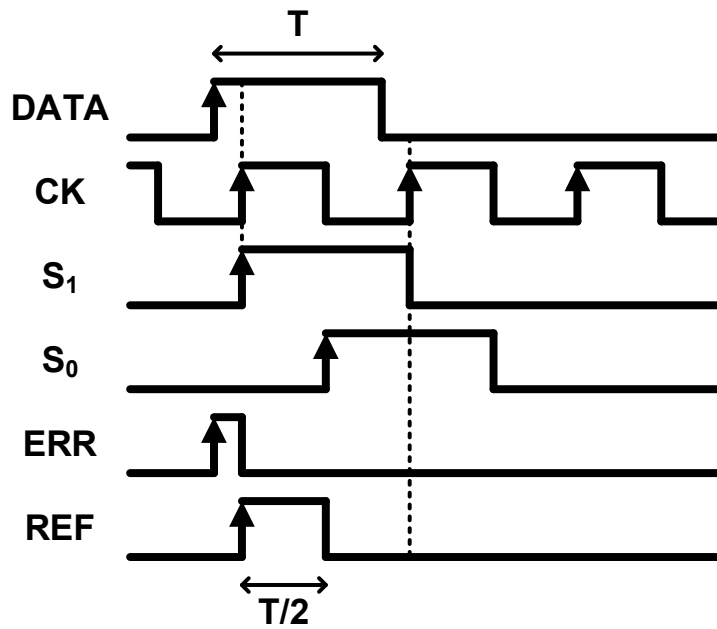
- A linear PD generates error signal proportional to input phase error
- The width of ERR is proportional to the phase error
- The width of REF is constant which is half clock cycle wide



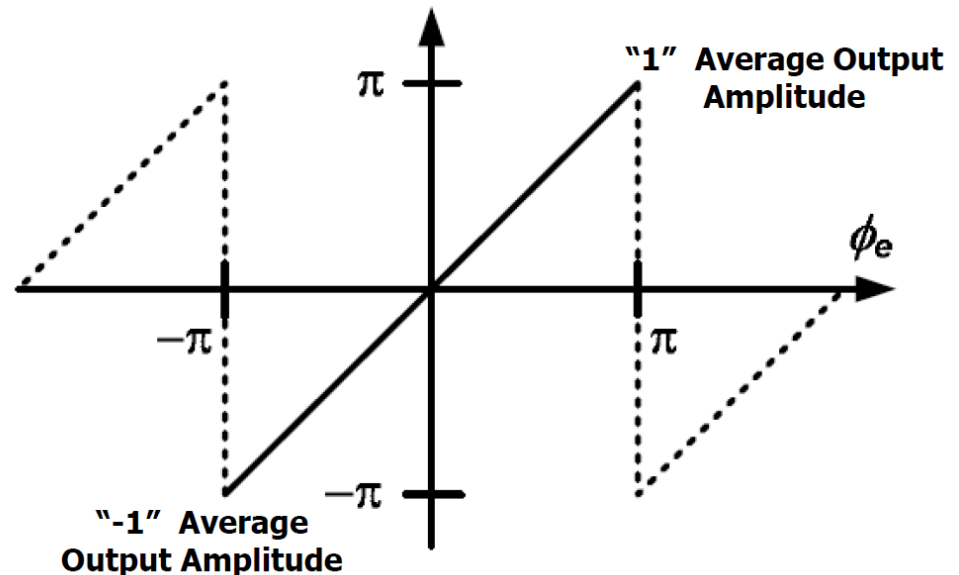
CDR Phase Detectors – Hogge PD

- Hogge phase detector

- A linear PD generates error signal proportional to input phase error
- The width of ERR is proportional to the phase error
- The width of REF is constant which is half clock cycle wide
- If clock is early (late), the width of ERR is smaller (larger) than the width of REF



Operation Principle

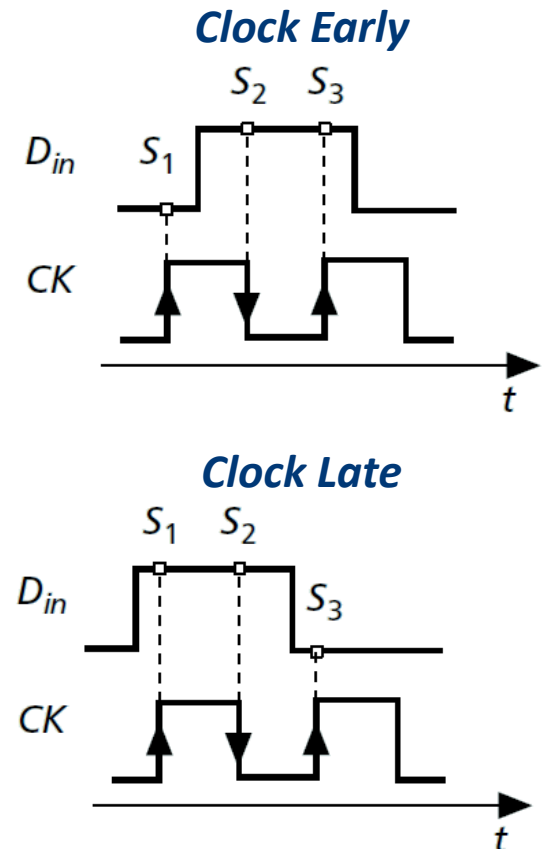
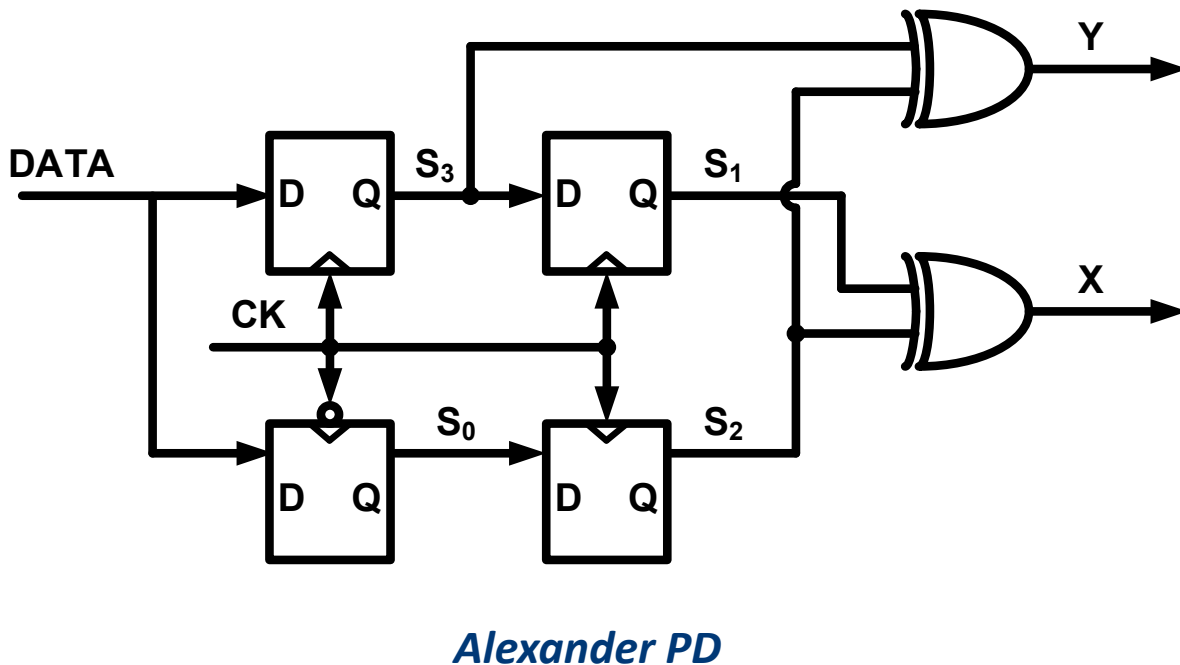


PD Gain Curve

CDR Phase Detectors – Alexander PD

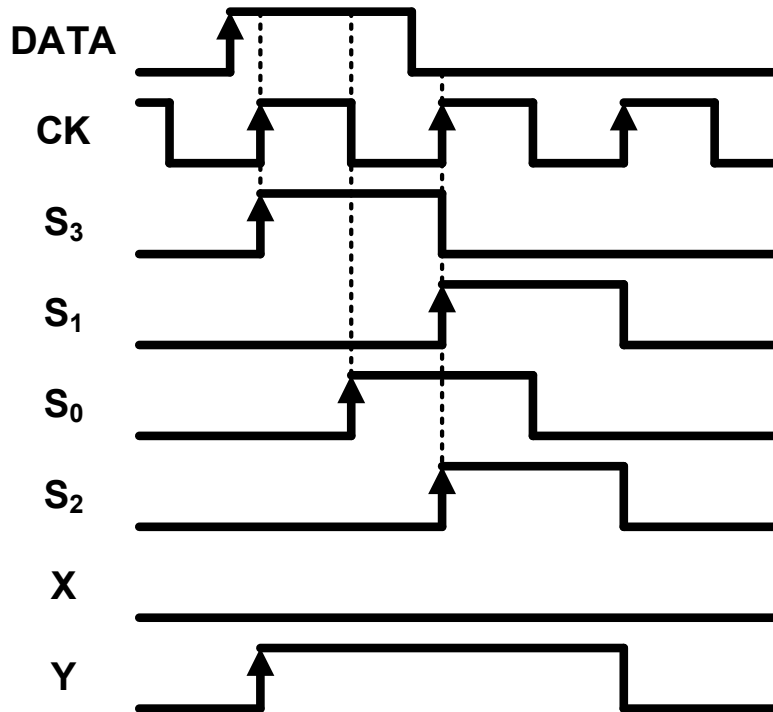
- Alexander phase detector

- Binary PD, Bang-bang PD (BBPD), 2x oversampling PD
- A binary PD detects only the sign of the phase error
- Simple and appropriate to high-speed operation

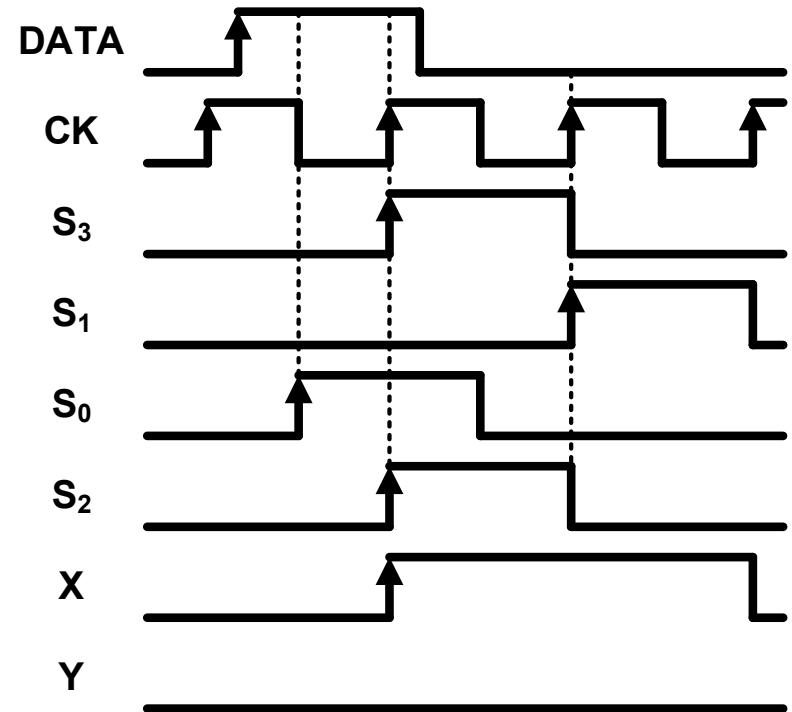


CDR Phase Detectors – Alexander PD

- Alexander phase detector
 - Bang-bang PD (BBPD), 2x oversampling PD
 - A binary PD detects only the sign of the phase error
 - Simple and appropriate to high-speed operation



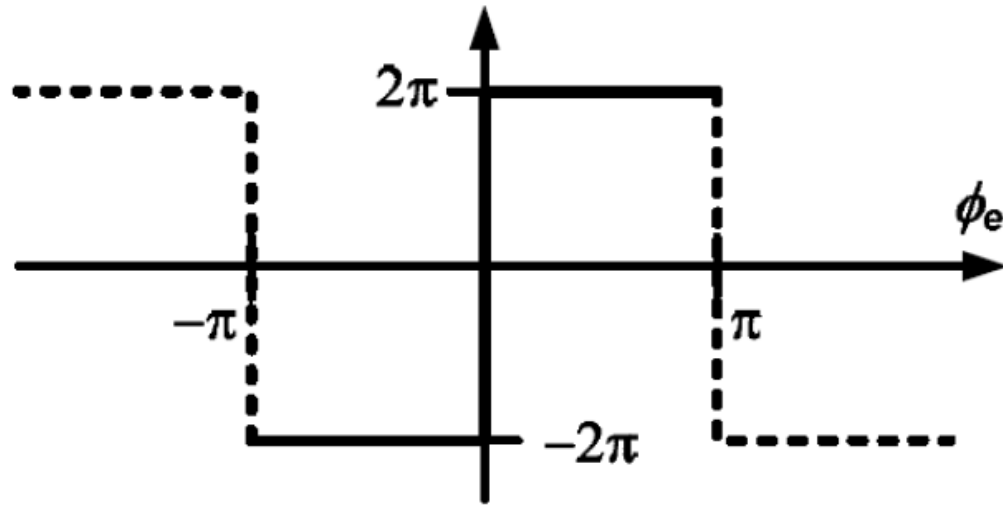
(a) Clock Late



(b) Clock Early

CDR Phase Detectors – Alexander PD

- Alexander phase detector
 - Bang-bang PD (BBPD), 2x oversampling PD
 - A binary PD detects only the sign of the phase error
 - Simple and appropriate to high-speed operation

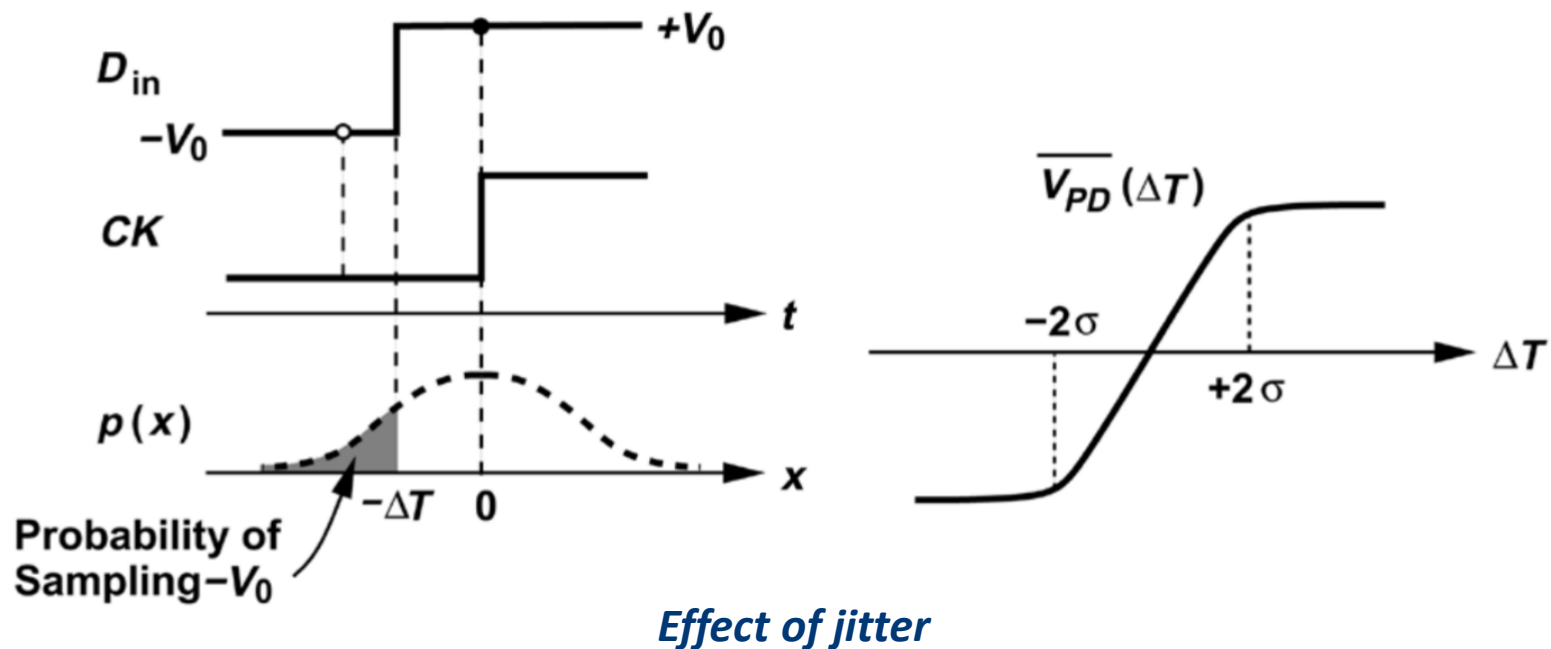


BBPD Gain Curve

CDR Phase Detectors – Alexander PD

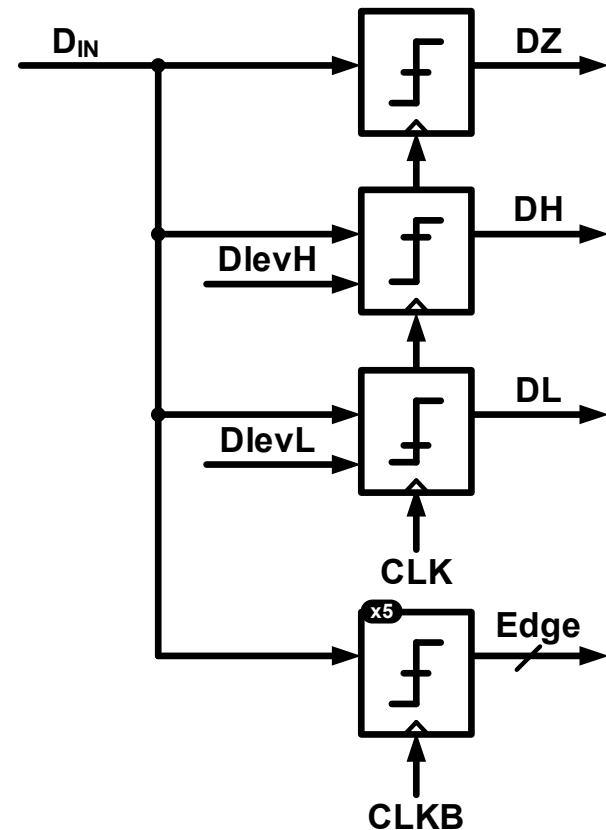
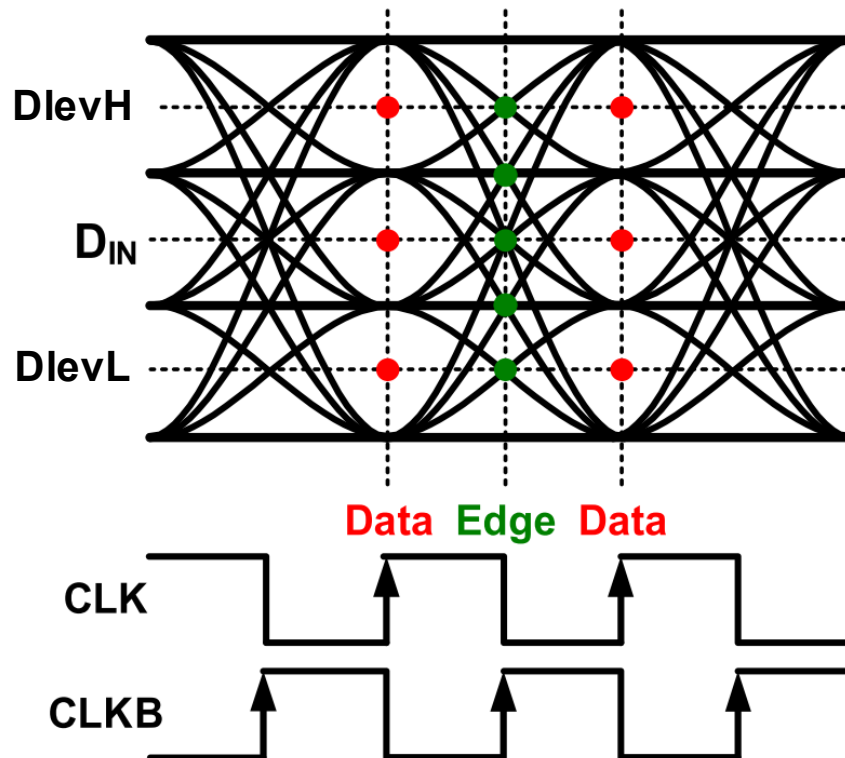
- Alexander phase detector

- The BBPD characteristic is smoothed out by the clock jitter
- The PD gain at small phase difference is inversely proportional to the standard deviations of the random jitter



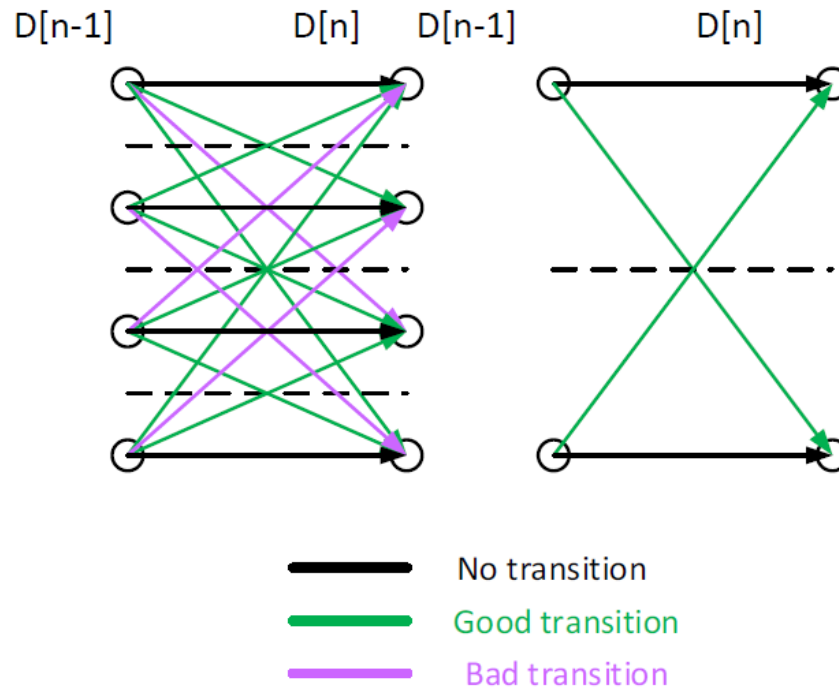
CDR in PAM4 Signaling

- In PAM4 signaling, **three threshold levels** for data sampling are needed
- Five edge samplers can obtain the full transitions in 2x oversampling
 - Half-rate clocking → total 6 data samplers and 10 edge samplers



PAM4 CDR Phase Detectors

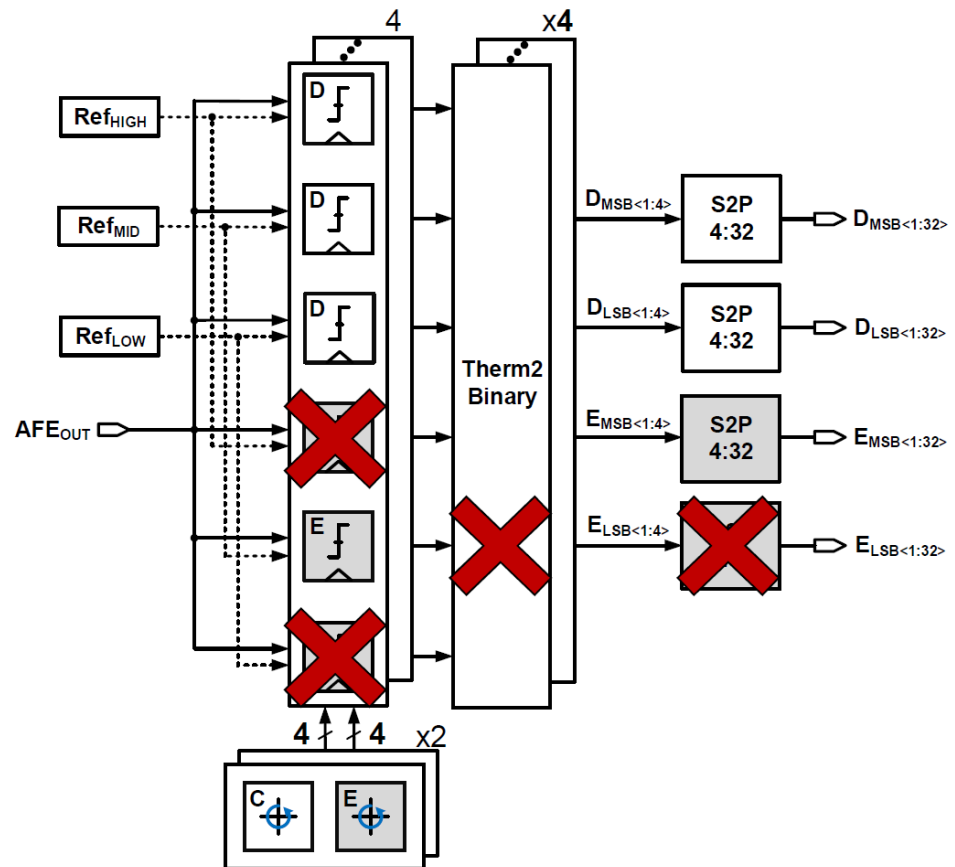
- PAM4 CDR using 2x oversampling
 - In NRZ, transition density is 50%
 - In PAM4, transition density depends on the number of edge samples
 - ✓ 5 edge samples → 75%
 - ✓ **3 edge samples → 50%, same transition density with NRZ**
 - ✓ 1 edge samples → 25%



PAM4 CDR Phase Detectors

- **PAM4 CDR using 2x oversampling**

- In NRZ, transition density is 50%
- In PAM4, transition density depends on the number of edge samples
 - ✓ 5 edge samples → 75%
 - ✓ 3 edge samples → 50%
 - ✓ 1 edge sample → 25%
- 3 edge samples are large overhead in high-speed link
- **1 edge sample !**
- How to compensate the reduced transition density?



PAM4 CDR Phase Detectors

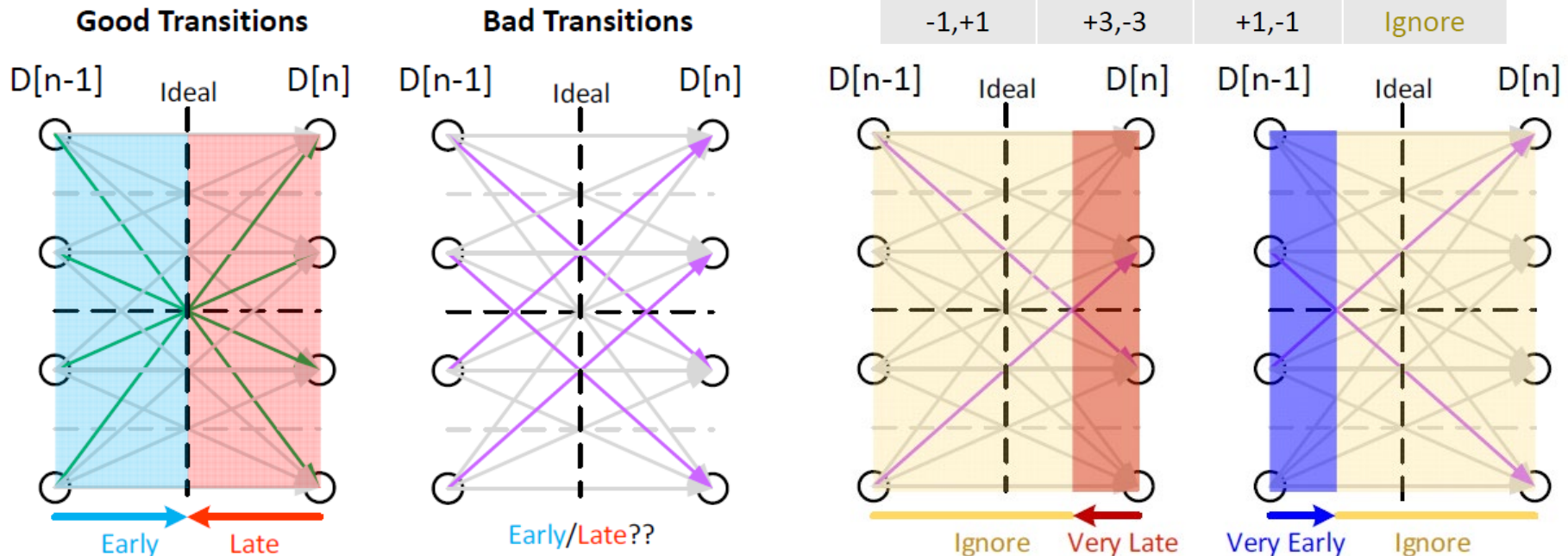
- PAM4 CDR using 2x oversampling**

- Utilizes bad transitions to generate very early and very late signals

D[n-1]	D[n]	E[n-1]	Early/Late
-3/-1/+1/+3	+3/+1/-1/-3	+1	Late
-3/-1/+1/+3	+3/+1/-1/-3	-1	Early

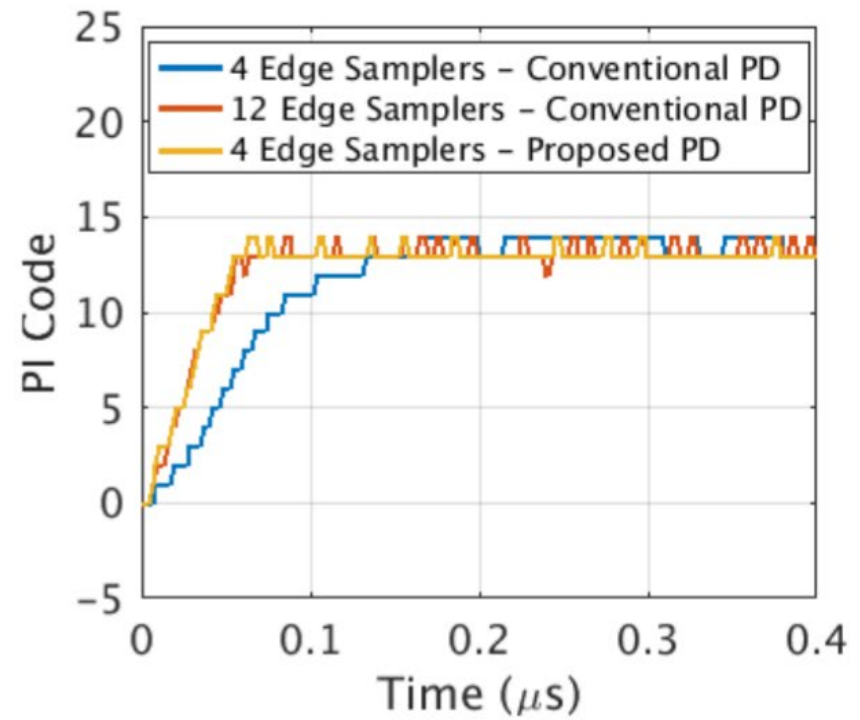
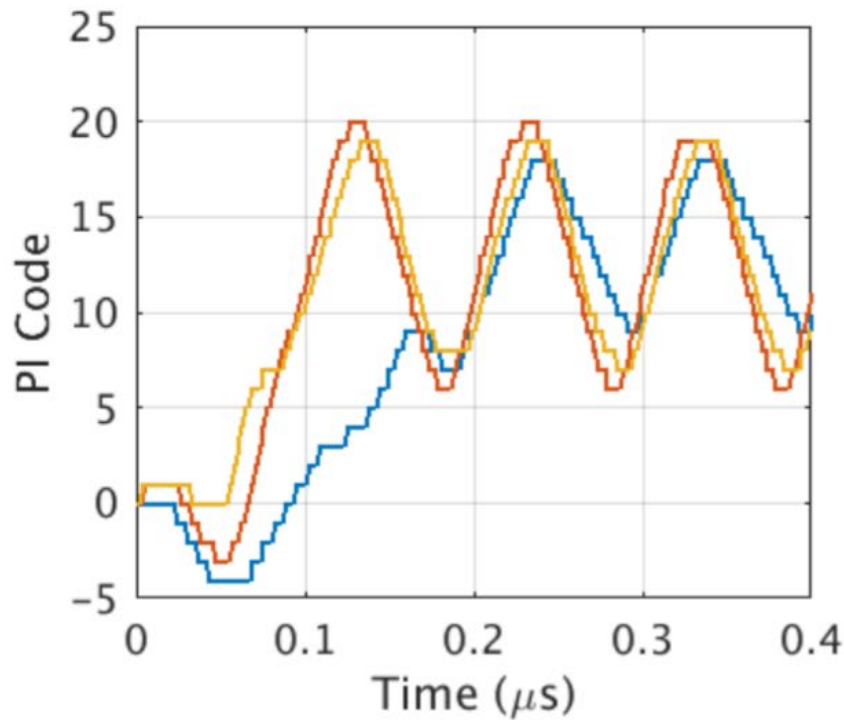
D[n-1]	D[n]	E[n-1]	Early/Late
-3,+3	+1,-1	+1,-1	Very Late
-3,+3	+1,-1	-1,+1	Ignore

D[n-1]	D[n]	E[n-1]	Early/Late
-1,+1	+3,-3	-1,+1	Very Early
-1,+1	+3,-3	+1,-1	Ignore



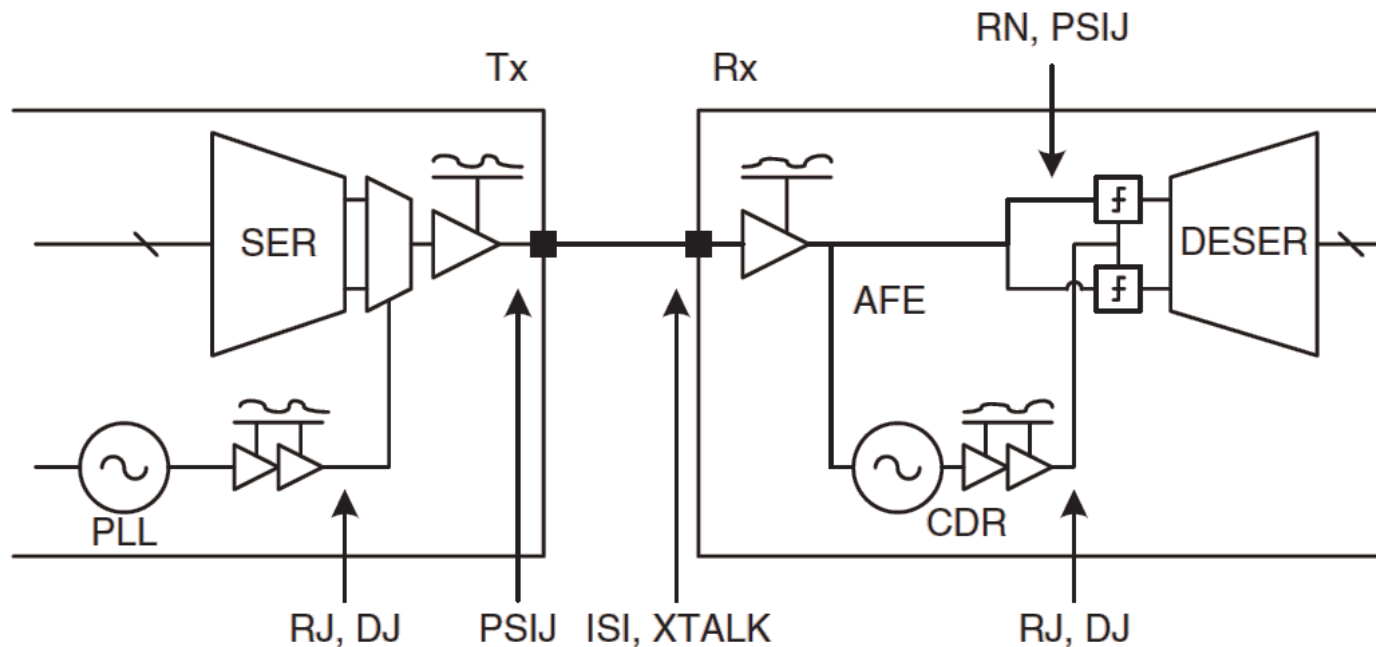
PAM4 CDR Phase Detectors

- **PAM4 CDR using 2x oversampling**
 - Utilizes bad transitions to generate very early and very late signals
 - Similar performance even using small number of samplers



Jitter in High-Speed Links

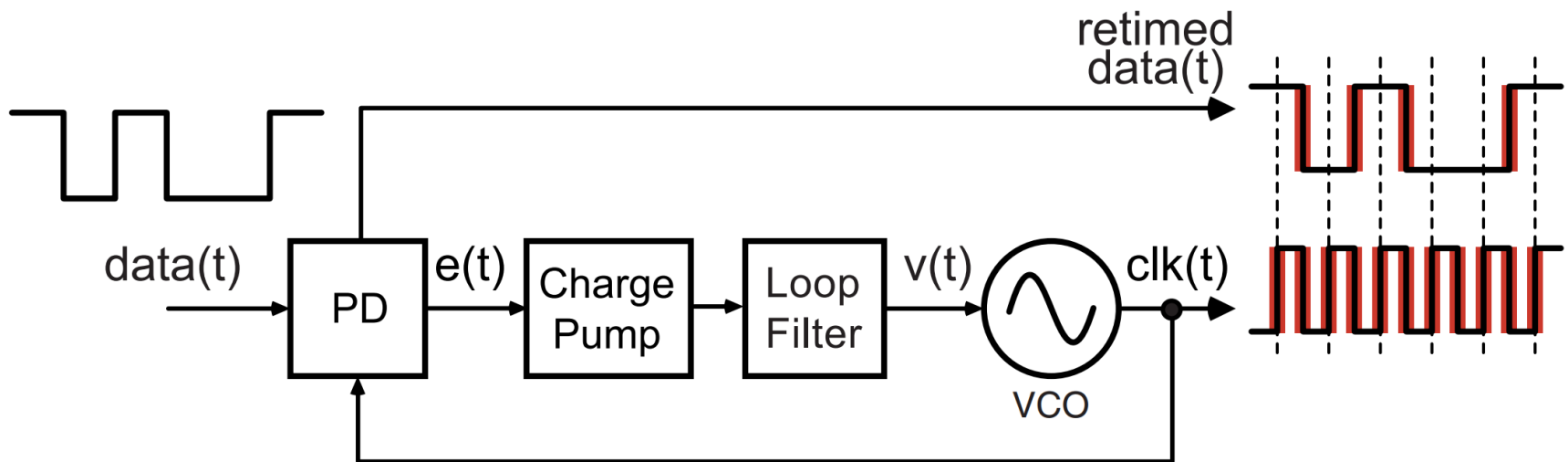
- Various jitter sources in view of receiver side
 - Data-dependent jitter: induced from lossy channel (ISI or crosstalk)
 - Source jitter: RJ and DJ from clocking circuits and PSIJ
 - CDR jitter: generated from clock recovery blocks such as VCO



RJ: Random Jitter DJ: Deterministic Jitter PSIJ: Power-supply-induced Jitter
ISI: Intersymbol Interference XTALK: Crosstalk

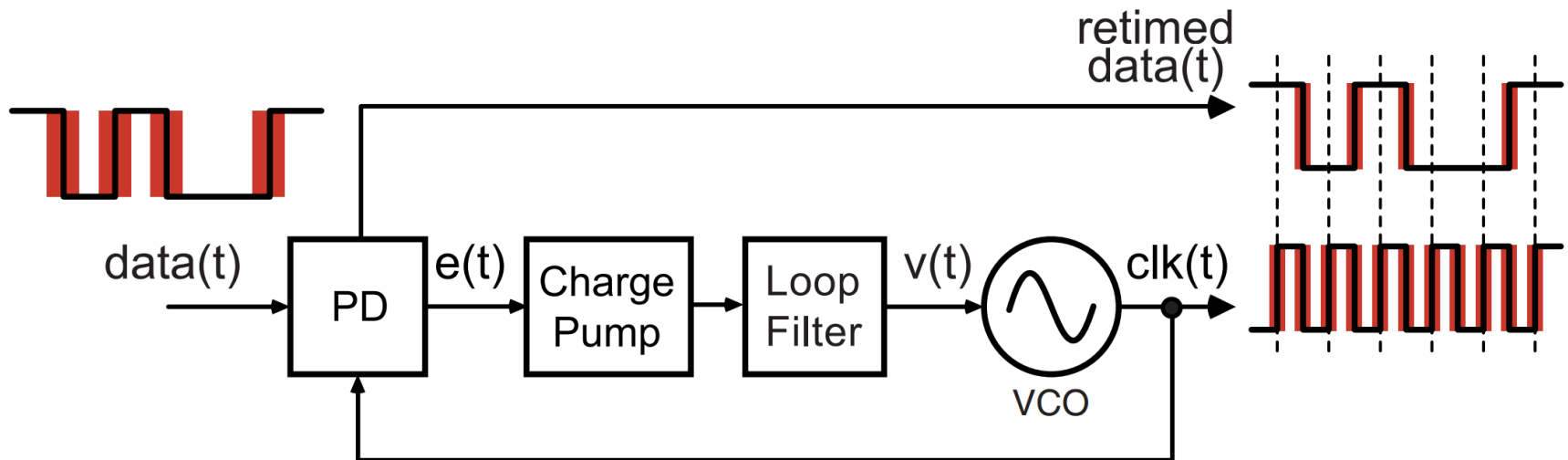
CDR Jitter Metrics – Jitter Generation

- Jitter generation (JGEN)
 - Definition
 - ✓ CDR output jitter measured with jitter-free input data
 - Jitter **generated** by CDR loop
 - SONET requires $JGEN < 10mUI_{rms}$



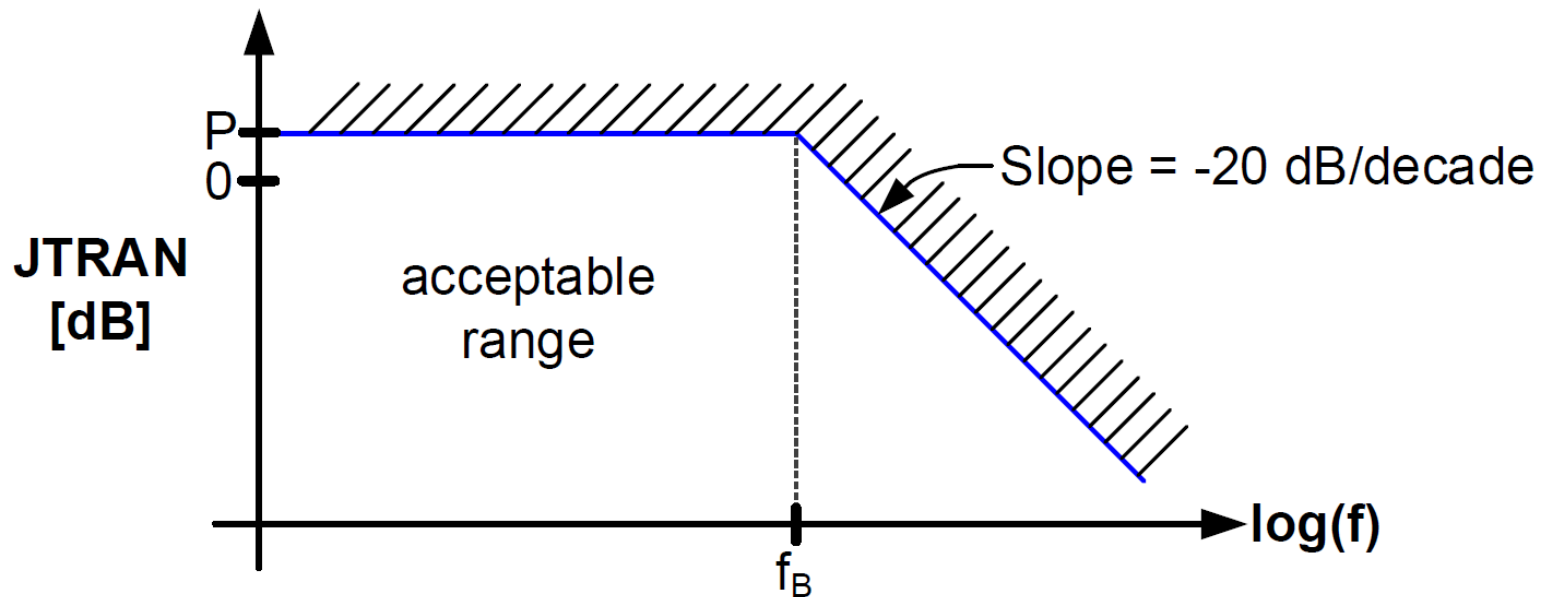
CDR Jitter Metrics – Jitter Transfer

- Jitter transfer (JTRAN)
 - Definition
 - ✓ Ratio of CDR output jitter to input sinusoidal jitter
 - Jitter **transferred** from input to output
 - Specified by jitter transfer bandwidth and jitter peaking



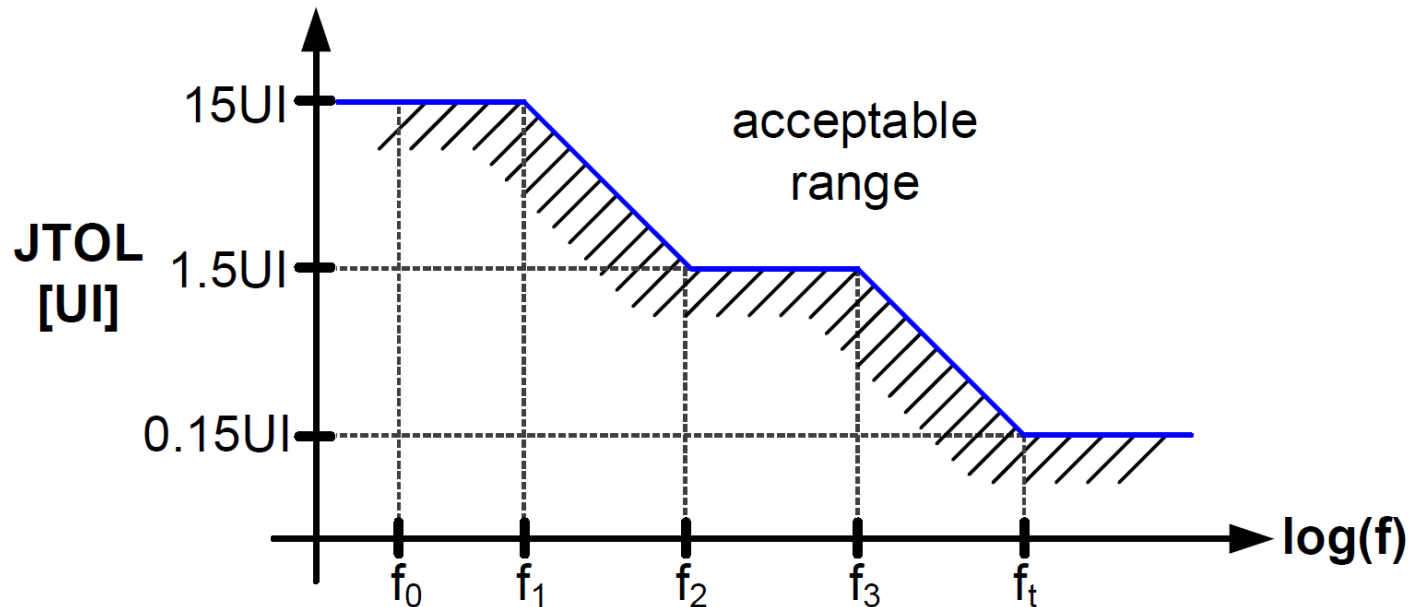
CDR Jitter Metrics – Jitter Transfer

- Jitter transfer (JTRAN)
 - Definition
 - ✓ Ratio of CDR output jitter to input sinusoidal jitter
 - Jitter **transferred** from input to output
 - Specified by jitter transfer bandwidth and jitter peaking
 - JTRAN mask: jitter peaking < 0.1dB for SONET OC-192



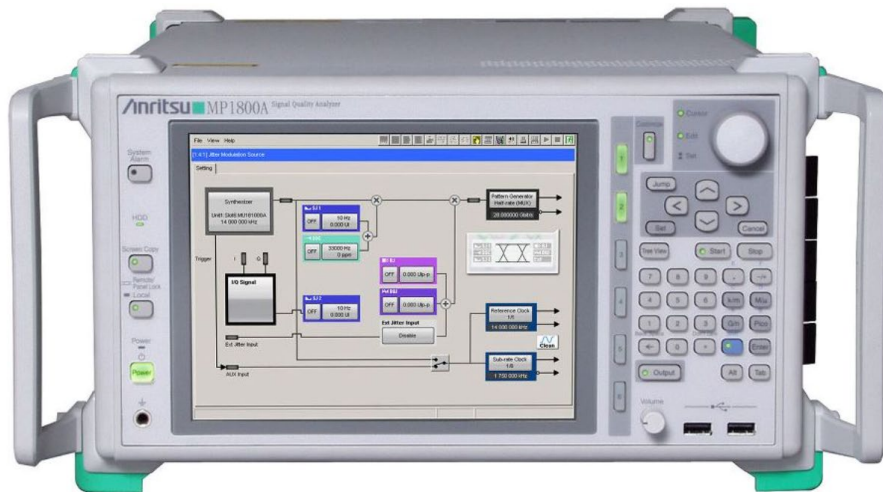
CDR Jitter Metrics – Jitter Tolerance

- Jitter tolerance (JTOL)
 - Definition
 - ✓ Maximum amount of jitter allowed on the input while still satisfying BER requirement
 - JTOL determined by CDR tracking ability
 - JTOL mask

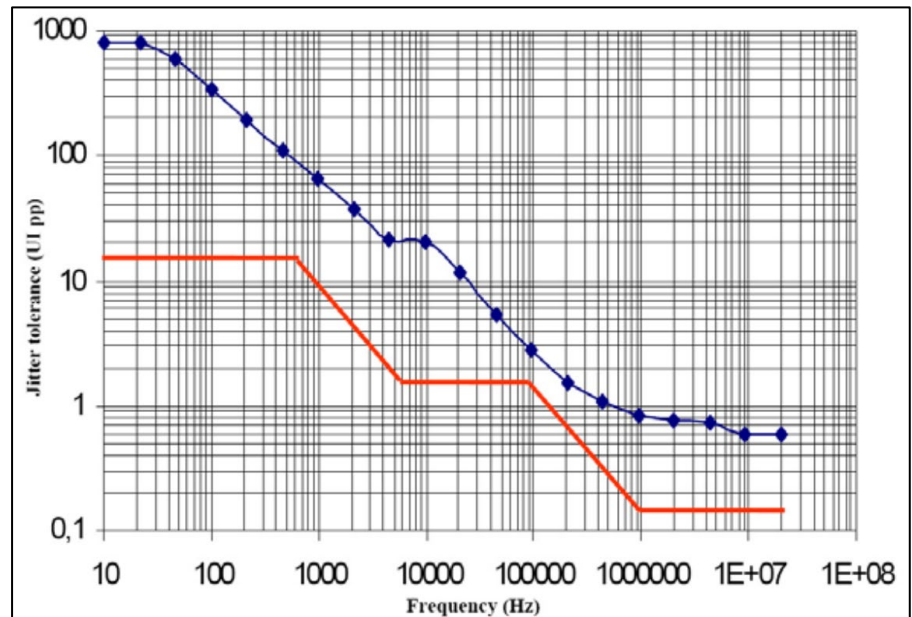


CDR Jitter Metrics – Jitter Tolerance

- Jitter tolerance (JTOL)
 - Measured by BERT



Anritsu MP1800A



Thank you !

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